

Monod: model-based discovery and integration through fitting stochastic transcriptional dynamics to single-cell sequencing data

Received: 2 May 2023

Accepted: 19 August 2025

Published online: 7 November 2025

 Check for updates

Gennady Gorin¹, Tara Chari², Maria Carilli³, John J. Vastola⁴ & Lior Pachter^{3,5}✉

Single-cell RNA sequencing analysis centers on illuminating cell diversity and understanding the transcriptional mechanisms underlying cellular function. These datasets are large, noisy and complex. Current analyses prioritize noise removal and dimensionality reduction to tackle these challenges and extract biological insight. We propose an alternative, physical approach to leverage the stochasticity, size and multimodal nature of these data to explicitly distinguish their biological and technical facets while revealing the underlying regulatory processes. With the Python package Monod, we demonstrate how nascent and mature RNA counts, present in most published datasets, can be meaningfully ‘integrated’ under biophysical models of transcription. By using variation in these modalities, we can identify transcriptional modulation not discernible through changes in average gene expression, quantitatively compare mechanistic hypotheses of gene regulation, analyze transcriptional data from different technologies within a common framework and minimize the use of opaque or distortive normalization and transformation techniques.

Technical advances in single-cell transcriptomics have enabled genome-wide molecular measurements in tens of thousands to millions of cells across diverse biological systems, from tissue samples and embryo models to whole organisms^{1–5}. To address challenges related to the processing and interpretation of the associated high-dimensional and noisy datasets, researchers using single-cell genomics analyses have settled on an amalgamation of heuristic data normalization, transformation and dimensionality reduction algorithms^{6–8}, which on the one hand extend bulk RNA sequencing (RNA-seq) analyses, and on the other hand adapt large-scale machine learning approaches to remove presumed technical noise and extract biological signal. Although these approaches are frequently employed, they have increasingly

recognized caveats related to hyperparameter sensitivity^{9–11}, distortion or loss of biological signal^{12–15}, incompatibility of qualitative methods with traditional statistical hypothesis-testing approaches^{9,15,16}, and interpretability^{16,17} that may require more attention as the scale and diversity of single-cell datasets grow to include features like chromatin state and protein counts^{13,18}.

A promising complementary approach, which respects the omnipresence of noise in the low copy number regime^{19–22}, is to fit mechanistic (discrete, stochastic) models to single-cell data. These models are interpretable by construction, can be rigorously formulated in terms of generative models, are straightforwardly compatible with traditional statistical tools and can reduce the necessity of

¹Fauna Bio, Emeryville, CA, USA. ²Eric and Wendy Schmidt Center, Broad Institute of MIT and Harvard, Cambridge, MA, USA. ³Division of Biology and Biological Engineering, California Institute of Technology, Pasadena, CA, USA. ⁴Department of Neurobiology, Harvard Medical School, Boston, MA, USA. ⁵Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, CA, USA. ✉e-mail: lpachter@caltech.edu

hyperparameter tuning by making the relationships between biological assumptions and model parameters more explicit. Few large-scale fits of such models to single-cell data have been performed thus far, both due to technical issues only partially addressed by previous work^{23–25} and nontechnical issues related to descriptive modeling being more prevalent than mechanistic modeling in genomics²⁶. Interestingly, this model-based approach is far more typical in the related field of fluorescence transcriptomics^{19,27–30}.

Here we present the Python package Monod, which supports a model-based approach to single-cell genomics by automating biophysical model fitting and certain kinds of statistical testing (Fig. 1a). In a typical scenario, the user defines a data likelihood by specifying both assumptions about the underlying biology, such as whether transcription is bursty²⁷, and about the measurement process, such as whether transcripts from longer genes tend to be captured more often²⁴. Monod provides methods to efficiently fit this likelihood to large-scale data and hence recover biophysically interpretable parameters along with estimates of their uncertainty; it also provides methods to run downstream analysis on these parameters, including comparing different biological hypotheses and testing for significant differences between experimental conditions.

The model-based approach to single-cell genomics does not replace the standard approach but rather extends it in some ways and complements it in others. It extends it by improving rigor and suggesting a richer set of biologically interesting hypotheses (that is, models beyond Poisson and negative binomial) to compare to data, and complements it by supplying natural alternative quantities on which to run statistical comparisons and perform clustering. The purpose of Monod is twofold: to leverage prior biophysical knowledge to extract more and different information from the same dataset, and to provide a rigorous computational framework for such analyses, which can be extended far beyond the use cases we consider here (see Chari et al.³¹ for one such extension).

In the following, we illustrate the capabilities of Monod through a comprehensive description of its modeling principles and a series of vignettes, including analyses of nine single-cell datasets that span diverse application domains (cell culture, cancer, radiation treatment, development, blood, and brain) and two species (mouse and human). We show that Monod can use nascent and mature messenger RNA (mRNA) count data to generate quantitative hypotheses about DNA and RNA regulation in heterogeneous cell populations, identify or assess model misspecification, integrate biology across sequencing platforms and techniques, and limit data distortion (Fig. 1a). Thus, as we continue to improve the depth and diversity of single-cell, molecular measurements, Monod provides a principled statistical foundation to represent relevant biological phenomena and questions about mechanisms of transcriptional variability, and to falsify these models and hypotheses based on the data. Further details about the Monod workflow are defined below and in ‘Overview of Monod inference’, with a visual representation and example outputs from the workflow shown in Supplementary Fig. 1.

Results

Assumptions and approach of Monod

Monod takes as input measurements of nascent and mature RNA count data, which can be obtained by aligning single-cell RNA-seq (scRNA-seq) reads to intron- and exon-containing references^{32,33}, and fits biophysical models of transcription. An advantage of the model-based approach is that it allows one to incorporate prior knowledge into analyses, such as the fact that the production of nascent and mature RNA is causally linked. Common bioinformatic analysis pipelines often treat nascent and mature RNA as independent rather than causally linked, conflate them or ignore nascent mRNA entirely^{13,33}.

All Monod models have two components: a biological component, which implements prior knowledge about the underlying biophysics,

and a technical component, which implements prior knowledge about how the measurement process tends to distort the underlying biology. Monod supports a variety of model types, but we will focus on models with the following structure:

$$\emptyset \xrightarrow{k} \mathcal{X}_N \xrightarrow{\beta} \mathcal{X}_M \xrightarrow{\gamma} \emptyset \quad (1)$$

$$\mathcal{X}_N \xrightarrow{C_N L} \mathcal{X}_N + \mathcal{X}'_N \quad \mathcal{X}_M \xrightarrow{\lambda_M} \mathcal{X}_M + \mathcal{X}'_M. \quad (2)$$

The underlying biological assumptions are that nascent RNA $\mathcal{X}_N$ is produced (with k the transcription rate), converted into mature RNA $\mathcal{X}_M$ (with β the splicing rate), and degraded (with γ the degradation rate). The underlying technical assumption is that a given mRNA molecule may be captured and reverse transcribed into cDNA multiple times, with the capture rate $C_N L$ of nascent RNA proportional to the gene length L and the capture rate of mature RNA independent of gene length^{24,25}. Monod assumes that input data are approximately at steady state; therefore, it fits all rate parameters in units of k , which is equivalent to imposing $k = 1$.

Monod supports a variety of biophysically plausible transcription models, each of which makes slightly different assumptions about transcription, splicing and degradation. This flexibility allows users to compare how well different biological hypotheses explain a given dataset; in general, one does not expect any single model to always perform best.

The simplest assumption one can make is that the probability per unit time of each process occurring is constant or, equivalently, that the waiting time associated with each process is exponentially distributed. If one assumes this for each process, one obtains the constitutive model, which yields Poisson RNA count distributions^{34,35} that do not exhibit the overdispersion commonly observed in real RNA count datasets^{27,36}. There are two well-known ways to complexify the constitutive model to incorporate overdispersion: the first is to assume that RNA is produced in ‘bursts’, with the burst size B assumed to be geometrically distributed^{27,37–43} (the bursty model); the second is to assume that the transcription rate k is gamma-distributed due to cell–cell heterogeneity^{23,44} (the extrinsic model). Each approach only modifies the first arrow in the biological component of the above model schematic.

Monod currently supports six models of transcription, including the constitutive, bursty and extrinsic models. The quasi-bursty limit of the Cox–Ingersoll–Ross model²³ and models that replace the exponentially distributed waiting times of either degradation or splicing with deterministic waiting times^{45,46} are also supported. We recommend the bursty model as the default choice for most analyses, as it is consistent with known biophysics and empirically fits many single-cell datasets well. See ‘Model selection and insight about gene regulation strategies’ for more discussion of each model type and how they compare in practice, and see the Supplementary Methods for the detailed definitions and properties of different models of biological and technical variability.

The inference process involves estimating two types of model parameters: biological parameters like β and γ , and the technical parameters C_N and λ_M . Because the technical parameters are only weakly identifiable²⁴, Monod assumes that they are shared across genes (but not necessarily across cell types). Monod estimates the technical parameters through a grid search and, conditional on a given set of technical parameters, estimates biological parameters via gradient descent on the Kullback–Leibler divergence (KLD) between the empirical and fitted distributions. Once a set of parameters that minimize the aforementioned divergence are identified, Monod quantifies parameter estimate uncertainty by exploiting the Fisher information matrix (FIM) of the data–model divergence. Monod also supports post hoc tests of goodness of fit, which it accomplishes by a combination of

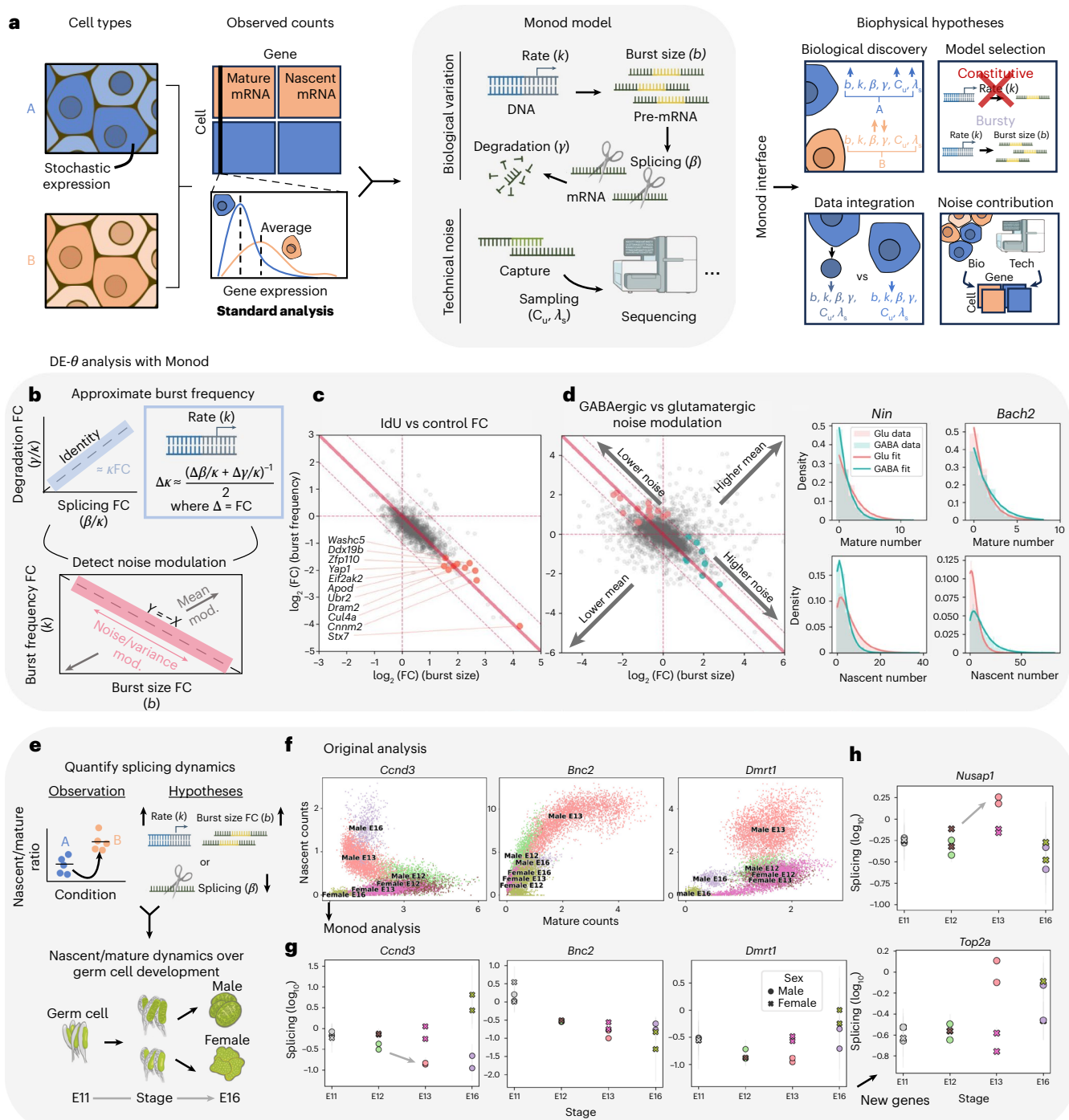


Fig. 1 | Monod inference for biophysical representation. **a**, Overview of the Monod framework for modeling multivariate genomics data. Monod infers biophysical parameters from joint count distributions to develop mechanistic hypotheses (example model of bursty transcription). **b**, Top, how changes in burst frequency (k) are calculated when splicing and degradation rate FCs are near identity. Bottom, changes in burst size and frequency denote noise versus mean expression modulation (mod.). **c**, Perturbation by IdU, which triggers DNA damage and repair, rarely changes expression levels but induces genome-wide noise enhancement⁵⁴ detectable by Monod (lines and gray points, as in **a**; red points, well-fit, moderate-expression genes identified as highly noise enhanced). **d**, Left, differences between mouse glutamatergic and GABAergic cell types ($n = 4$ replicates) include genes with substantial noise but little to no change in average expression. Light red (teal) points, significantly higher noise in glutamatergic (GABAergic) cells; gray points, all other genes; solid diagonal line, parameters for which burst size and frequency compensate to maintain a constant average expression; dashed diagonal lines, $\pm 1 \log_2$ (expression FC)

region about the constant average expression line; vertical and horizontal lines, parameter combinations for which burst size and frequency, respectively, do not change. Right, cell type-specific distribution shapes (light red, glutamatergic cell type; light teal, GABAergic cell type; histograms, raw counts; lines, Monod fits; top, mature RNA marginal; bottom, nascent RNA marginal). **e**, How changes in ratios of nascent and mature mRNA counts can derive from different mechanistic hypotheses. Analyses with Monod were carried out with a developmental dataset of mouse germ cells (male and female). E, embryonic day. **f**, Nascent and mature counts of key genes in the original study⁵⁷, over embryo stages and sexes. **g**, Monod splicing rate fits for these key genes for each of the $n = 2$ biological replicates. Error bars, 99% confidence regions determined by $\pm 2.576\sigma$. The gray arrow indicates changing splicing rates. **h**, Trends for new candidate genes for which the splicing rate increases over the stages in testis germ cells. The gray arrow indicates changing splicing rates. Images in **a** adapted from NIAID NIH BIOART (<https://bioart.niaid.nih.gov>); panels **e** and **f** adapted from ref. ⁵⁷ under a Creative Commons license (CC BY 4.0).

criteria based on χ^2 statistics, Hellinger distances from the theoretical distribution and adequate distance of parameters from user-imposed bounds (see example rejected genes in Supplementary Fig. 3). For more on the inference process, see ‘Overview of Monod inference’ and previous work²⁴.

Given an assumed model, Monod’s output is a set of model parameter estimates, parameter uncertainties and goodness-of-fit metrics (see example outputs in Supplementary Figs. 1 and 3). In the following vignettes, we discuss some mechanistic questions that can be interrogated using Monod fits, essentially by comparing parameter fits between biological samples or models.

Monod generalizes differential expression analyses

The goal of single-cell genomics studies is often to perform exploratory analysis of complex biological systems to develop new questions and hypotheses about how different cell states and types are defined and result in heterogeneous functions. Common approaches use distance metrics or averages of gene expression across cells to identify cell clusters and ‘neighbors’, and compare expression profiles across conditions and continuous processes. Using biophysical inference, we can generalize this assessment of cell type heterogeneity beyond a descriptive, data scientific comparison of summary statistics and representation of cells as points in average gene expression space. The representation of cells through their biophysical characteristics provides a more general definition of cell ‘type’, simultaneously summarizing the molecule distributions and more closely encoding the intuition that cells comprise, regulate and are exposed to stochastically varying environments^{47,48}.

In typical transcriptomic workflows, determination of differences between cell types or states relies on identification of differentially expressed (DE) genes^{49–51}, which exhibit statistically significant differences in their average gene counts. However, these differences can be defined by both changes in mean and variance of gene expression, and regulatory mechanisms such as modulation of transcriptional noise^{47,48} can maintain a constant mean and would not be identifiable by standard statistical methods. The identification of differential expression is also impacted by treatment of technical covariates⁶. Thus, the mechanistic Monod framework generalizes the notion of differential behavior to modulated parameters, which describe not only how the distribution of gene counts changes but also which underlying biological processes effect this change.

Here we demonstrate how differential parameter behavior captures alterations to the mean and variance of gene expression counts as well as quantifies changes in specific biological processes such as pre-mRNA splicing that would otherwise rely on more qualitative analyses. This extension of DE to biophysical parameters is denoted as DE- θ , where θ may be a data moment or an inferred biophysical parameter (Fig. 1b).

Modulation of transcriptional noise, which may or may not result in changes to average gene expression, has been shown to affect cell fate determination⁵² and DNA damage recovery, among other processes⁴⁷, leading to growing interest in identifying noise regulators in the cell⁵³. To demonstrate the ability of Monod DE- θ to detect modulations without substantial changes in average expression, we first applied Monod to data from a recent study⁴⁷, which showed that the introduction of a modified nucleotide (5'-iodo-2'-deoxyuridine, IdU) to culture medium enhances transcriptional noise but keeps average expression constant, hinting at a genome-wide mechanism for compensation^{47,54} (Fig. 1b,c). The inferred parameters for the bursty model with length-biased capture demonstrated striking noise amplification in IdU-perturbed cells, with limited differences in mean expression (Fig. 1c). This included genes that generally encode mediators of the cellular stress response: *Zfp110*, *Eif2ak2* and *Yap1* encode regulators of apoptosis; *Dram2*, involved in the autophagic response to DNA damage repair; *Cul4a*, involved in the turnover of DNA repair proteins; and *Ddx19b*, which

encodes a product potentially active in stress granules (Fig. 1c). These findings are consistent with the authors’ conclusions and independent reanalysis, which likewise found that burst size increases and burst frequency decreases in the IdU condition^{47,54,55}. The ascribed change in burst frequency reflects the change in inferred splicing and degradation rates being near identity, as it is more likely that such coordinated changes in unrelated processes are a result of changes to the denominator (k) (‘Signatures of frequency modulation’).

We also found that noise modulation is present not only in this more artificial perturbation context but also in heterogeneous cell types, such as between glutamatergic and GABAergic neurons in the mouse brain⁵⁶ (Fig. 1d). For example, *Nin* and *Bach2*, the products of which are involved in neuronal development, visually exhibited higher noise in glutamatergic and GABAergic populations, respectively (Fig. 1d). The mature count averages were, on the other hand, fairly close (*Nin* (glutamatergic, 1.7; GABAergic, 0.98), *Bach2* (glutamatergic, 0.87; GABAergic, 1.4)). Separately from burst modulation, splicing rate dynamics are also of interest, particularly from data containing nascent and mature mRNA counts, to, for example, understand splicing kinetics over germ cell development and maturation⁵⁷ (Fig. 1e). Previous analysis inferred changes in splicing by observed changes in the nascent/mature mRNA ratio (Fig. 1f). However, multiple alterations to transcription as well as splicing mechanisms can result in such ratio differences (Fig. 1e). With Monod-inferred parameters, we could identify when the splicing rate itself was likely modulated, which we applied to genes in the original study, the spermatogenesis-related genes *Bnc2* and *Dmrt1*, as well as the cell cycle-related gene *Ccnd3* (ref. 57). We found greater differences in splicing rate between the ovary and testis germ cells in the E13 stage for *Ccnd3* and *Dmrt1* as posited by Mayère et al.⁵⁷ (Fig. 1g). Given these parameters, we also identified new gene candidates that displayed enhanced, rather than reduced, splicing kinetics during development. These include the mitotic spindle-associated gene *Nusap1* and the topoisomerase-encoding gene *Top2a* with roles in transcription and DNA replication⁵⁸, which show increased splicing rates between the E12 and E13 stages (Fig. 1h).

Monod identifies strategies of resistance and recovery

The expanded capabilities of biophysical DE analysis also allow us to explore more complex biological settings, in which it is less clear which mechanisms are expected to change and which genes provide actionable targets for follow-up experiments. For example, in disease contexts, it is important to understand how aberrant behaviors in tumor or cancer cell populations promote tumorigenesis and drug resistance to develop targeted therapeutics and interventions⁵⁹. Likewise, understanding cellular strategies of recovery after treatment aids in the selection of relevant therapies and the promotion of desired cell behaviors⁶⁰.

To first explore how cells may promote undesirable behaviors such as drug resistance and malignancy, we analyzed patient samples of pancreatic ductal adenocarcinoma (PDAC) tumors. Single-nucleus RNA-seq (snRNA-seq) was performed to improve retention of malignant cell populations in patient samples with and without neoadjuvant treatment⁶¹ (Fig. 2a). We analyzed treated samples from patients who received multicycle chemotherapy (FOLFIRINOX) followed by multi-fraction radiotherapy with concurrent fluorouracil (5-FU)–capecitabine (CRT)⁶¹ or CRT and losartan (CRTI) treatment (Fig. 2a). Given the variability in patient backgrounds, we compared the biophysical parameters inferred for a CRT (and CRTI) patient sample to the parameters of three untreated samples to identify robust changes for each treated patient. We compared malignant cells (given existing annotations⁶¹) across the tumor samples, and parameters were inferred for the bursty model of transcription with length-biased sampling over a shared set of 3,000 genes, with 1,030 genes retained based on goodness-of-fit metrics across samples.

As identified in the original publication⁶¹, CRT- and CRTI-treated tumors had higher mean expression (where μ_M are scaled pseudobulk

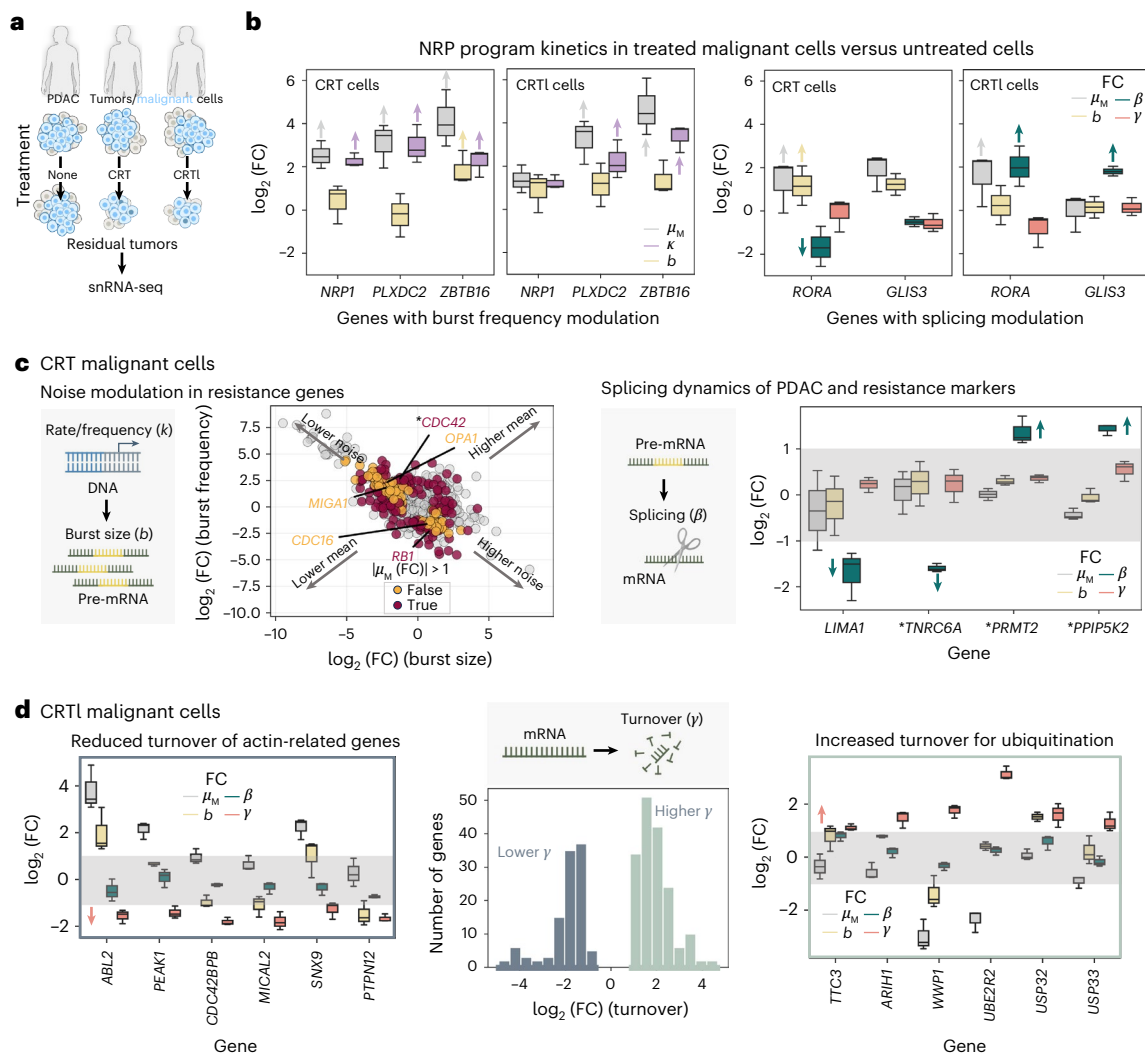


Fig. 2 | Resistance and proliferation in malignant PDAC cells. a, Diagram of snRNA-seq data from residual PDAC tumors⁶¹ after treatment (or untreated, as in first column), obtained from frozen patient samples. **b**, Left, box plots showing $\log_2(\text{FC})$ of mean (mature) mRNA expression alongside burst size and burst frequency FC, between the denoted condition and untreated samples ($n = 3$). Whiskers denote $1.5 \times$ the interquartile range, for which the lower boundary is the minimum value and the upper boundary is the maximum value, excluding outliers. Right, box plots showing $\log_2(\text{FC})$ of mean (mature) mRNA expression alongside burst size, splicing rate and degradation rate between the denoted condition and untreated samples ($n = 3$). Whiskers denote $1.5 \times$ the interquartile range, for which the lower boundary is the minimum value and the upper boundary is the maximum value, excluding outliers. **c**, Left, plot of $\log_2(\text{FC})$ in burst frequency versus burst size for all nonrejected genes in malignant cells treated with CRT. Genes with $\log_2(\text{FC}) > 1$ in both or either parameter are colored. Red denotes mean expression $|\log_2(\text{FC})| > 1$, and yellow denotes $|\log_2(\text{FC})| < 1$.

(little or no change). Right, box plots showing $\log_2(\text{FC})$ of mean (mature) mRNA expression alongside burst size, splicing rate and degradation rate between cells treated with CRT and untreated samples ($n = 3$). Whiskers denote $1.5 \times$ the interquartile range, for which the lower boundary is the minimum value and the upper boundary is the maximum value, excluding outliers. *Published genes associated with 5-FU resistance or chemoresistance. Gray box denotes $|\log_2(\text{FC})| < 1$. **d**, Left, box plots showing $\log_2(\text{FC})$ of mean (mature) mRNA expression alongside burst size, splicing rate and degradation rate between cells treated with CRTI and untreated samples ($n = 3$). Whiskers denote $1.5 \times$ the interquartile range, for which the lower boundary is the minimum value and the upper boundary is the maximum value, excluding outliers. Gray box denotes $|\log_2(\text{FC})| < 1$. Middle, histogram of genes with reduced turnover ($\log_2(\text{FC}) < -1$) or increased turnover ($\log_2(\text{FC}) > 1$). Right, same box plot as left for genes with increased turnover. Images in **a** adapted from NIAID NIH BIOART (<https://bioart.niaid.nih.gov>).

mature mRNA counts) of genes belonging to a neural-like progenitor malignant program than the untreated samples (Fig. 2b and 'Data processing: PDAC tumor samples'). These genes are related to neuronal development, migration and adhesion, which can increase invasiveness and lead to poor prognoses⁶¹. In addition to mean changes, we also identified which processes were likely modulated, denoted by the arrows in Fig. 2b. Burst frequency (or transcription rate k) modulation was determined as described above ('Signatures of frequency modulation') (Fig. 2b, left), distinct from scenarios in which splicing rate specifically had the greatest contribution to the observed fold change (FC) in mean expression (Fig. 2b, right).

We can additionally identify genes, not covered in previous analyses, that specifically demonstrate transcription-based modulation, as opposed to splicing- or degradation-based modulation, potentially through noise regulation. In CRT-treated malignant cells, the tumor suppressor gene *RBI* (ref. 62) showed distinct downregulation through burst frequency (transcription rate), while *CDC42*, the product of which regulates cell division and can aid resistance in 5-FU-treated colorectal cancers⁶³, demonstrated increased burst frequency (Fig. 2c, left). A set of genes related to mitochondrial fusion and spindle organization, *MIGA1*, *OPA1* and *CDC16* (ref. 58), did not have discernible changes in mean but did demonstrate modulation of burst frequency and/or size,

suggesting strategies to increase or decrease their transcriptional noise (Fig. 2c, left). Their modulation echoes previous literature, which has described the enrichment of genes related to organelle fusion and fission and spindle structure in 5-FU-resistant colorectal cancer cells⁶⁴. Beyond transcriptional control, we also found marked splicing rate changes in genes associated with drug resistance or tumorigenesis, for which mean expression differences may be low. *TNRC6A*⁶⁵, *PIPSK2* (ref. 66) and *PRMT2* (a known PDAC marker)⁶⁷ all represent genes associated with 5-FU resistance and/or chemoresistance (Fig. 2c, right). In particular, increases in *PIPSK2* and *PRMT2* have been associated with increased drug resistance^{66,67}, with both genes also showing increased splicing rates. Decreased splicing was also observed for *LIMA1*, for which lower expression has been associated with increased tumorigenicity⁶⁸. This presents new gene candidates for which splicing regulation, in addition to transcriptional machinery, is a potential regulatory step for enacting therapeutic control⁵⁹.

In CRTI-treated malignant cells, we can similarly identify changes in noise modulation, for example, with decreasing noise for *TIMP2*, previously found to be regulated by losartan treatment⁶⁹, and increased noise for the proto-oncogene *MET*⁷⁰ (Supplementary Fig. 4). Notably, we can also extract genes for which mRNA turnover is uniquely affected. We refer to the learned degradation rate as general ‘turnover’, as snRNA-seq may better capture nuclear export as opposed to decay or degradation. Within the set of genes with decreased or slower turnover, we found genes related to regulation of actin–cytoskeletal dynamics encoding actin-associated kinases (*ABL2*, *PEAK1* and *CDC42BPB*) as well as *MICAL2* and *SNX9* (related to actin depolymerization and polymerization, respectively)⁵⁸ and *PTPN12* (related to actin reorganization and a potential tumor suppressor⁷¹) (Fig. 2d, left). Such turnover-based strategies have been similarly observed in proliferating cells, in which β -actin mRNA turnover decreased with faster growth rates⁷¹, and in tumor cells in which actin-binding proteins can accumulate in the nucleus promoting development of invasive structures⁷². In the opposite direction, we found that genes encoding E3 ubiquitin–protein ligases (*TTC3*, *ARIH1* and *WWP1*) and ubiquitin–conjugating enzymes and deubiquitinating enzymes (*UBE2R2*, *USP33* and *USP32*)⁵⁸ had increased turnover and/or export, suggesting potential control of protein ubiquitination and/or turnover by altered mRNA turnover (Fig. 2d, right). Together, these results also present gene candidates regulated through their export and stability that can promote resistance and proliferation of these malignant cells.

In addition to understanding how cells develop tumorigenic and resistance properties, exploring their ability to compensate and recover from treatment is also important to enable more effective therapies and treatment plans. For example, the intestine is one of the organs most sensitive to radiation damage: understanding how it heals after radiotherapy is important for promoting healthy recovery and survival of patients treated for abdominal, gastrointestinal, pelvic and retroperitoneal tumors⁶⁰. Particularly interesting is how immune cells are affected by and respond to radiation treatment. T cells, which are essential actors in the body’s natural response to tumor control^{73,74}, are required for successful radiotherapy but are also believed to be damaged during treatment. While studies have investigated the transcriptomic landscapes of T cells after radiotherapy^{60,73}, we suggest that a fuller understanding of the regulation of key genes, as revealed by biophysical parameter changes, could inform how therapeutics may be designed and more efficiently administered to enhance recovery from radiotherapy.

We reanalyzed data from a study of the transcriptomic response of the intestinal microenvironment to radiation-induced intestinal injury (RIII) in wild-type C57BL/6J male mice⁶⁰ (Fig. 3a). Mice were treated with 15 Gy of abdominal radiation, and intestinal samples were collected before radiation treatment (day 0) and at several time points after treatment (days 1, 3, 7 and 14 after treatment). We selected T cells based on the annotations of the original study and inferred biophysical

parameters for samples (each sample represents a unique mouse) with at least 100 T cells, leaving 11 samples across days 0, 1, 3 and 14, with at least two samples per day. We used the bursty model with Poisson technical sampling noise, and, after fitting 3,000 genes shared across samples, subset to the 1,182 that passed goodness-of-fit testing for all samples, further correcting inferred parameters for sample-specific offsets (‘Data processing: irradiated T cells’).

We first show that analysis through the lens of a biophysical model reveals more genes with differences in behavior over the time course of recovery. To assess the significance of parameter changes, we performed a t-test with unequal variance on the $\log_{10}$ of each corrected parameter per gene for day 0 versus all other days. Specifically, we tested for changes in burst size (b), relative splicing rate (β), relative degradation rate (γ) and normalized mature counts (μ_m ; ‘Data processing: irradiated T cells’), correcting P values for the number of tested genes using the Benjamini–Hochberg procedure (separately for each parameter) to obtain false discovery rates (FDRs). Fig. 3b shows that many genes that do not show significant differences in mature counts are detectable via parameter changes: testing the significance of inferred parameters in addition to means allows for the exploration of more genes for which significant differences between days or conditions may lie in regulatory strategy rather than mean expression change. For example, comparing day 1 to day 0, there were 380 genes with significant differences in a parameter but not in mature expression, 127 genes with differences in both and 157 genes with significant differences in mature expression but not parameters (Fig. 3b). We further emphasize that the analysis of parameters does not preclude testing for changes in mean, with our method adding to the list of possible genes for follow-up investigation.

In addition to identifying genes with parameter changes that are not detectable using standard mature expression comparisons, for genes that do show significant differences in expression, Monod can suggest which cellular regulatory strategy is responsible (for example, an increase in burst size or a decrease in degradation rate; Fig. 3c). For example, we show in Fig. 3c (left) five genes that had significant increases between day 1 and day 0 in both burst size and mature RNA counts. As these genes also show an increased degradation rate, the observed increase in mature counts is likely due to an increase in burst size (that exceeds the increase in degradation rate). These genes, which are upregulated early in the radiation response time course (day 1 after treatment), have been studied as promising targets in cancer immunotherapy. *Cbl* encodes the E3 ubiquitin ligase Casitas B cell lymphoma, which regulates immune cells, and its homolog Casitas B cell lymphoma b (CBL-b) has been shown to promote immunosuppressive tumor microenvironments⁷⁵. *Nt5e* encodes 5′-nucleotidase (or CD73), an immunosuppressive protein for which high expression is also associated with poor outcome of the tumor microenvironments of patients with cancer⁷⁶ and is currently emerging as a novel target for immunotherapy⁷⁶. Other genes for which high expression has been correlated with lower overall survival of cancers are *Ncapd3*, which is associated with reduced infiltration of immune cells into glioma⁷⁷, and the gene that codes for insulin-like growth factor 2 receptor (*Igf2r*), for which the human equivalent was found to be an indicator of poor prognosis in patients with triple-negative breast cancer⁷⁸. On the other hand, the expression of *Mpp7*, which codes for a scaffolding protein, has been shown to have a positive correlation with increased infiltration of T cells in melanomas⁷⁹.

Other notable genes show increases in mature RNA expression best explicable by decreases in degradation rate (Fig. 3c, right). The proto-oncogene *Ets1* encodes a transcription factor that promotes aggressive tumor behavior in a variety of cell types⁸⁰. While not directly associated with cancer progression, the proteins encoded by *Pak2* and *Git2* are known to interact⁸¹ and are crucial for the stability and suppressive function of regulatory T cells⁸². With knowledge about what specific point in the RNA lifecycle (for example, bursty transcription

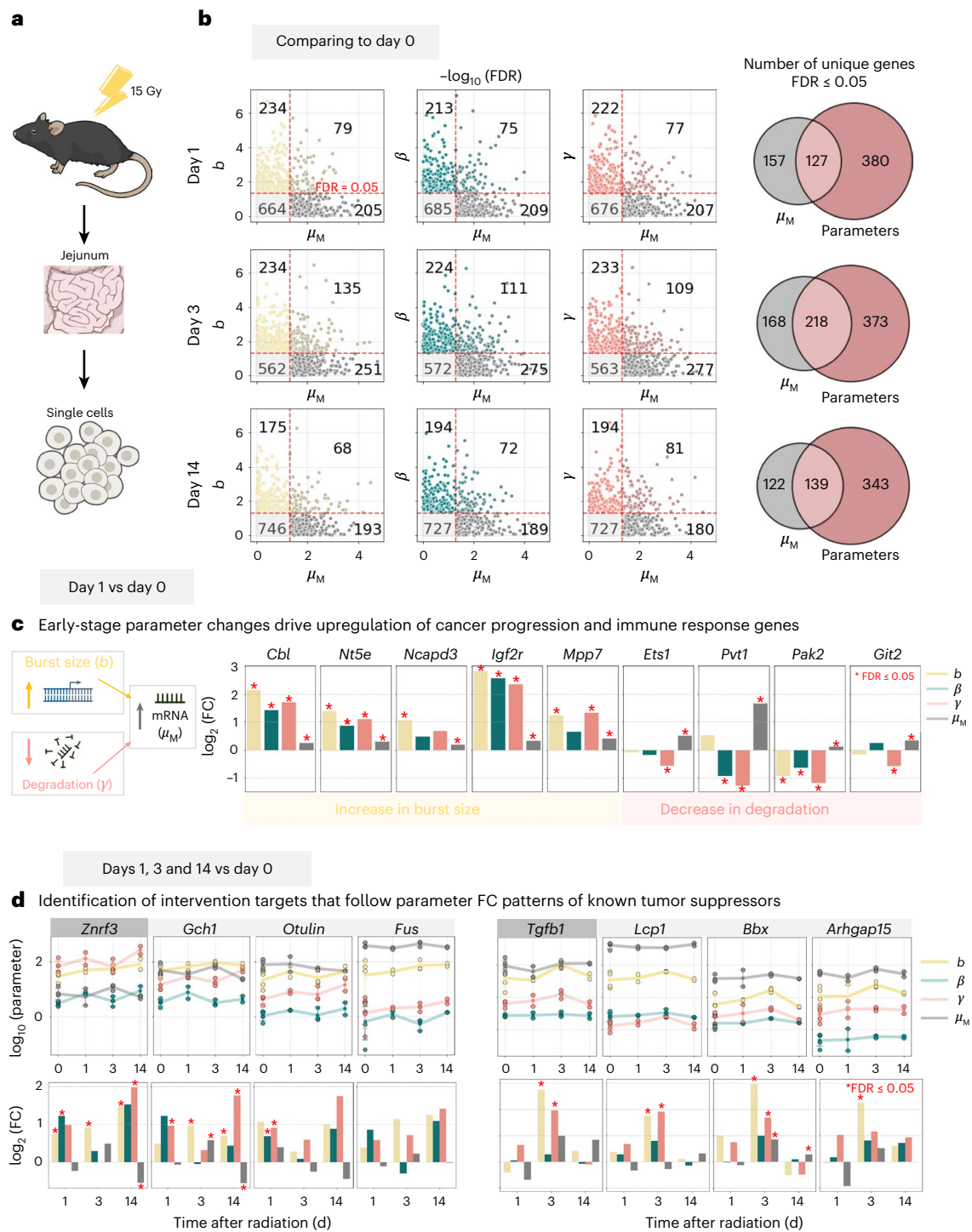


Fig. 3 | Monod analysis of mechanisms of T cell recovery after radiation treatment. **a**, Single-cell samples were taken from the intestines of mice exposed to 15 Gy of abdominal radiation before treatment and at several time points after treatment. **b**, Two-sided t -tests comparing T cells on days 1 ($n = 2$), 3 ($n = 3$) and 14 ($n = 2$) after treatment to T cells before treatment (day 0, $n = 4$) flag more genes as significant ($FDR \leq 0.05$) when comparing inferred parameters (b , β and γ) than when merely comparing mature RNA counts (μ_M). In the scatterplots, red dashed lines indicate a significance threshold of $FDR = 0.05$. **c**, The $\log_2$ (FC) values of parameters (comparing day 1 to day 0; 'Data processing: irradiated T cells') show that statistically significant changes in the mature mean expression of genes related to the progression of cancer and immune response can be

attributed to statistically significant differences in parameters (that is, burst size or degradation rate). Red asterisks indicate $FDR \leq 0.05$. **d**, Comparing gene parameter changes over the time course of recovery can identify similarly regulated genes. Top, $\log_{10}$ of parameters and mean mature RNA expression on days 0, 1, 3 and 14. Each outlined point indicates a separate sample (mouse) with a smaller point at the mean over the $\log_{10}$ of sample parameters per day. Error bars indicate the interquartile range of the sample parameters per day. For certain genes, the $\log_2$ (FC) of each parameter and mean mature expression for days 1, 3 and 14 from day 0 ('Data processing: irradiated T cells') exhibit patterns similar to those of known tumor suppressors (bottom). Images in **a** adapted from NIAID NIH BIOART (<https://bioart.niaid.nih.gov>).

or degradation) may lead to increased expression of a gene, researchers could design strategies to discourage known tumor promoters and encourage known tumor suppressors for increased efficacy and complementarity of radiotherapy and immunotherapy.

Finally, we can identify genes for which parameters show similar dynamics (FCs) over the time course of recovery from radiation treatment as compared to those of known tumor-suppressing genes ('Data processing: irradiated T cells'). For example, we found three genes with FC patterns most similar to that of *Znrf3* (Fig. 3), for which the encoded protein negatively influences cancer progression by inhibiting key proliferative (WNT) signaling pathways^{83,84}; and another three with patterns most similar to that of *Tgfb1*, the product of which potentially suppresses tumorigenesis at early cancer stages but promotes tumorigenesis at advanced cancer stages⁸⁴. Of these genes, some have been reported as important in cancer: for example, *Gch1* is a known immunosuppression gene with upregulation previously linked to reduced survival in breast cancer⁸⁵, *Lcp1* encodes a prognostic biomarker in gastric cancer⁸⁶ and *Arhgap15* codes for a protein that was reported to slow colorectal tumor growth⁸⁷. Other identified genes have not been as well characterized in cancer studies: *Otulin* encodes an immune response peptidase⁸⁸, and *Fus* and *Bbx* are more general DNA damage response genes^{89,90}. Analysis of biophysical parameters allows one to explore how genes that have been previously implicated in cancer prognoses experience similar regulatory changes in response to radiation treatment as well as to identify genes that have not been previously implicated as targets for follow-up therapeutic development.

Model selection and insight about gene regulation strategies

Monod currently supports six sophisticated yet tractable models of transcription, previously shown to describe and distinguish between features of experimental data^{23,25,45} for thousands of genes across multiple cell types. We emphasize that Monod supports modular model comparison, in the sense that one can freely modify assumptions about a specific model component (for example, whether degradation occurs with a deterministic or stochastic waiting time) to obtain a new model to test against data. We recommend using Monod with the bursty model by default, as it captures observed overdispersion³⁶ and known biophysics^{27,43} while only minimally complexifying the constitutive model. But the bursty model may not always fit the data best, and it is worth testing to what extent a less complex, more complex or simply different model performs better in practice.

In this section, we use Monod to analyze mouse brain data with 2,951 genes and six cell types (Allen sample B08 (ref. 56)) and show that the bursty model almost always outperforms the simpler constitutive model and tends to perform fairly well compared with alternative models (see Fig. 4a for schematics of the five models we consider in this section). Moreover, we show that it is useful to consider a richer space of transcription models, as specific models may fit specific genes or cell types better or worse, which suggests that Monod-derived fits yield insight about gene- and cell type-specific gene regulation strategies. We reject models (for a given cell type and gene) if they fail to pass the goodness-of-fit thresholds described in 'Biophysical model selection and rejection'.

We first compare the constitutive model with the bursty model. For many genes, only the bursty model passes goodness-of-fit thresholds. When neither the bursty model nor the constitutive models is rejected for a gene, the bursty model is usually a better fit according to the Akaike information criterion (AIC; 'Biophysical model selection and rejection' and Fig. 4b). It is extremely rare that the constitutive model fits the data better, perhaps unsurprisingly as the constitutive model is a special case of the bursty model and the overdispersion of single-cell count data is well known but diverging from previous studies that advocate the Poisson model⁹¹.

Although the transcriptional behavior of many genes is consistent with the bursty model, comparing all five of the (comparably complex)

proposed models reveals that the extrinsic, delayed degradation and delayed splicing models are better fits for some genes in different glutamatergic cell types (Fig. 4c). This comparison also suggests differences between cell types' transcriptional regulation strategies, with more genes best fit by the extrinsic model in layer 6 intratelencephalic (L6 IT) neurons than in layer 5/6 near-projecting (L5/6 NP) neurons, for example (Fig. 4c). Furthermore, including the additional models increases the number of genes in each cell type for which one or more models pass goodness-of-fit thresholds: fewer genes are rejected by all models when considering more models (Fig. 4d). As all models were compared in a standard manner using the same data, fitting pipeline and statistical rejection criteria, the discrepancies between proposed models could be evidence of real biological variety in transcriptional and post-transcriptional regulation mechanisms.

While cell type comparisons show distinctions between the number of genes best fit by each model in a given cell type, Monod also provides the tools to investigate how genes exhibit expression patterns consistent with different transcriptional models across cell types (Fig. 4e). We found the model with the largest AIC weight for every gene in every cell type and then, for each gene, collapsed models over cell types to a set of unique models. The UpSet plot in Fig. 4e shows how many genes exhibited each combination of models preferentially across the six cell types. For example, only three genes exhibited only bursty behavior preferentially, while 580 genes were best fit by the bursty model in at least one cell type and by the extrinsic model in at least one other cell type. All but 27 genes exhibited expression trends described best by at least two unique models, with the most common set (622 genes) being bursty, extrinsic and delayed degradation. Interestingly, no genes were best fit by only the delayed splicing model or only the constitutive model across the six cell types (Fig. 4g).

While these trends may reflect biological differences, they may also highlight differences in the mutual identifiability of the proposed models. For example, the bursty, extrinsic and delayed degradation models all account for correlation between nascent and mature counts, while, in the constitutive and delayed splicing models, nascent and mature counts are uncorrelated. Thus the genes that are best fit by the three models or some subset may in fact display similar behavior across different cell types, with the correlation between their nascent and mature counts, ruling out the constitutive and delayed splicing models, as the salient feature driving the fits (similar to Gorin et al.⁴⁵).

To explore this question, we can assess the identifiability of models or the extent to which the best-fit model is 'better' than other models. While we can assign a gene to the model with the largest AIC weight ('Biophysical model selection and rejection'), there is not always a clear distinction between model fits. To illustrate this, we show a simplex of normalized AIC weights for the bursty, extrinsic and delayed degradation models, colored by the assigned regulatory model: the weights for the best-fit models are sometimes arbitrarily close to the weights of the other two models (Fig. 4f). In this way, genes with clear preference for a given model can be distinguished from those that are similarly fit by two or more models. These results can also be verified by visualizing model-based probabilities and normalized empirical count histograms. Fig. 4g shows genes that were not rejected by only the bursty model (*Wipf2*), only the extrinsic model (*Ubl3*) and only the delayed degradation model (*Ildr2*) in layer 6 neurons (L6 CT). *Eprs1* and *Ufd1*, indicated with red stars in Fig. 4f, were fit similarly well under the three models. In the histograms for *Wipf2*, *Ubl3* and *Ildr2*, we highlight the (Poisson) constitutive model's poor ability to recapitulate the overdispersed data, consistent with Fig. 4b,c.

Monod makes fitting subtly distinct mechanistic models of transcription for thousands of genes straightforward, thereby leading to increased ease of model comparison and selection. Performing goodness-of-fit and likelihood tests on Monod fits provides a route for discerning between biophysical models that may have generated the data. However, we note that definitive model selection is beyond the

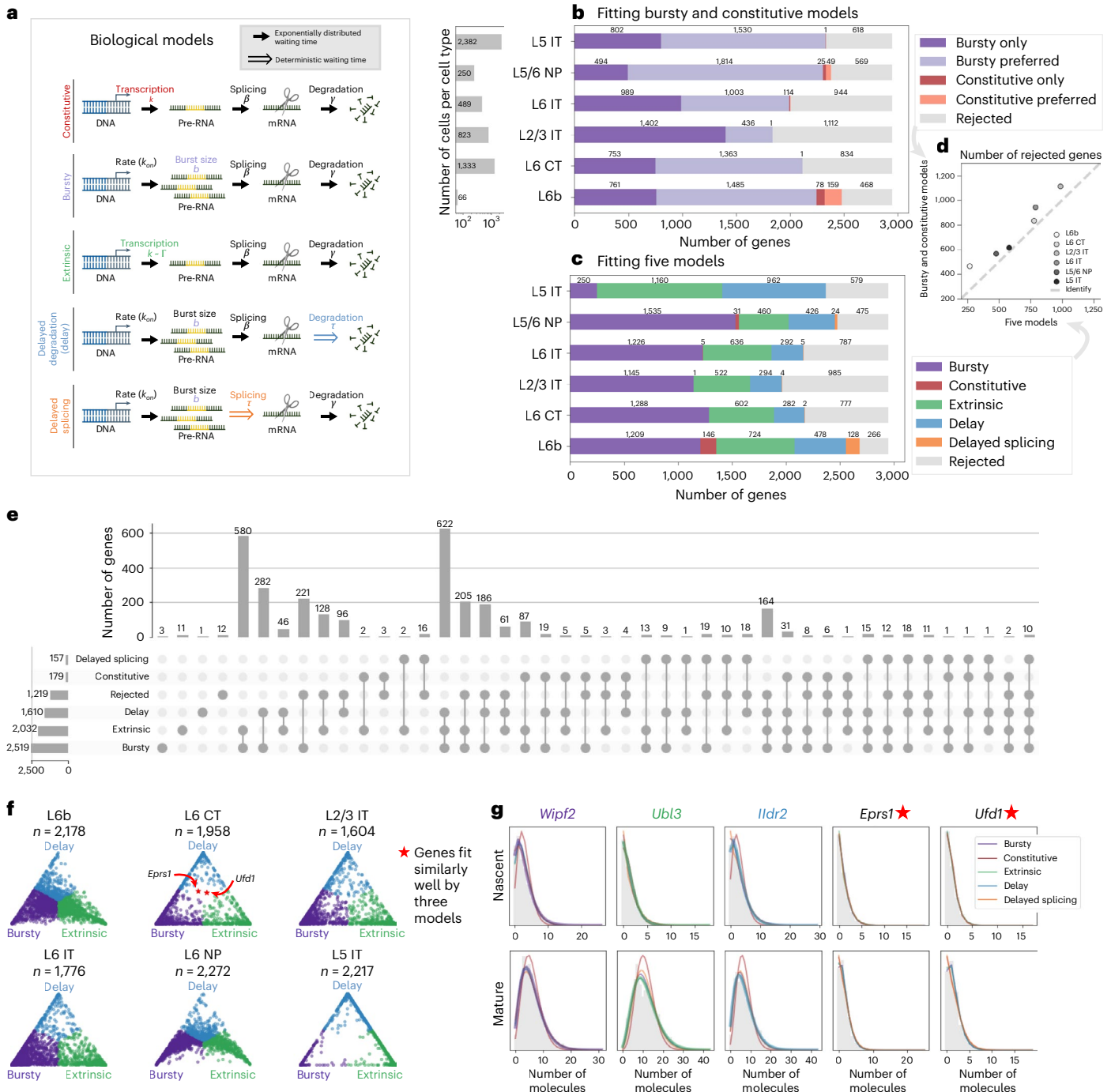


Fig. 4 | Monod facilitates the comparison of different biophysical models on a per-gene basis for thousands of genes across cell types. Results are shown for Allen sample B08 (ref. 56). **a**, Diagrams of five of the biological models implemented in Monod, varying in transcription and processing kinetics (Supplementary Methods (section S1.1)). **b**, Of the 2,951 fit genes in each of six cell types, most genes display behavior more consistent with the bursty model than with the constitutive model using the AIC and two goodness-of-fit metrics ('Biophysical model selection and rejection'). **c**, For the same six cell types and genes, the number of genes that are best fit by each of the five considered biological models, colored by model. Gray indicates genes for which all models were rejected. **d**, Fewer genes are rejected when fitting and comparing five models than when fitting and comparing only the bursty and constitutive

models. **e**, The same gene can exhibit different behavior in different cell types: the UpSet plot shows the number of genes that are best fit by a given model or combination of models across the six cell types. **f**, Ternary diagram of normalized AIC weights of genes for the three most commonly selected models, colored by the model with the largest AIC weight. Only genes for which none of these models were rejected are displayed, with the number indicated. L6b, L2/3 IT and L5 IT refer to layer 6b, layer 2/3 intratelencephalic and layer 5 intratelencephalic neurons, respectively. **g**, Predicted marginal nascent and mature distributions for each model plotted over normalized empirical histograms of molecule counts for the indicated genes in L6 CT neurons. *Eprs1* and *Ufd1*, which had similar AIC weights for the bursty, extrinsic and delay models, are noted with red stars in the ternary diagrams.

scope of Monod: rather than coming to definite conclusions, Monod should be used to explore data and to compare and generate hypotheses that can later be tested experimentally.

Principled integration of multiple modalities

How should one reconcile or combine scRNA-seq data with other types of single-cell data, such as snRNA-seq data^{92,93}, single-molecule

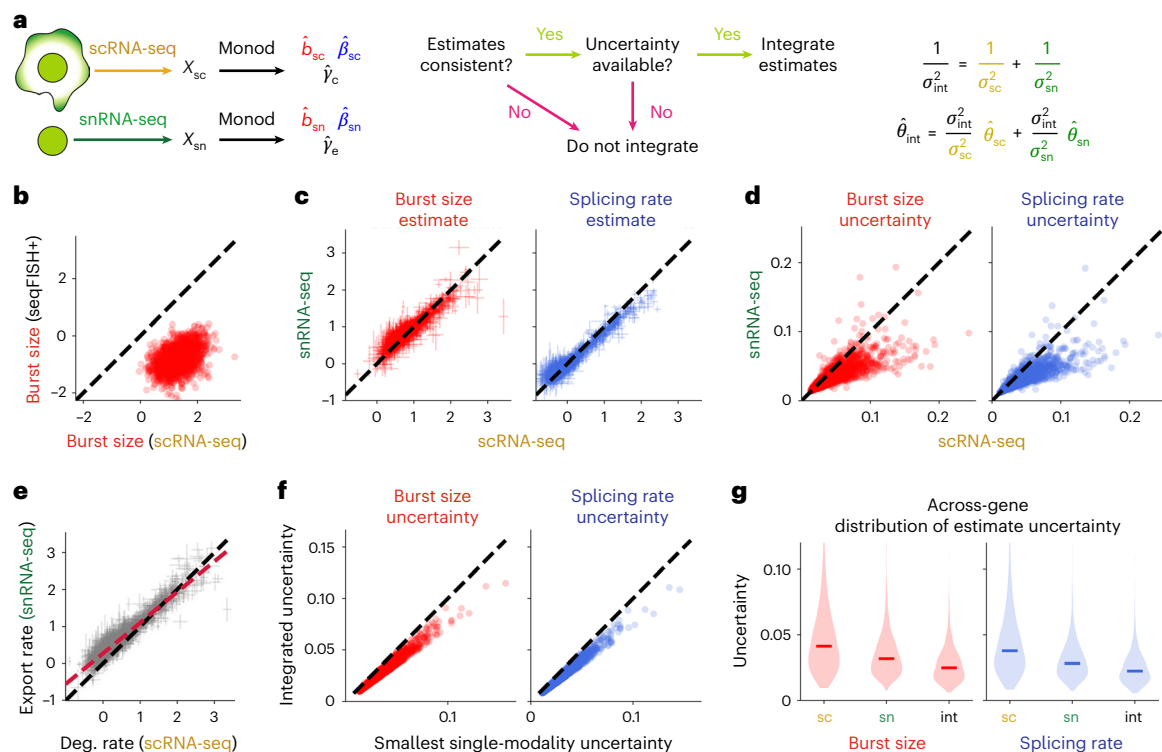


Fig. 5 | Monod's quantification of parameter estimate uncertainty supports a biophysically meaningful approach to integrating data from different modalities. **a**, Schematic depiction of integration (int), with fitting single-cell and single-nucleus data to the bursty model used as an example. Integrating data from different modalities (sc, scRNA-seq; sn, snRNA-seq) is useful (when it is possible) because it reduces the uncertainty of parameter estimates. **b**, Consistency of seqFISH+ and scRNA-seq burst size estimates for 1,964 genes in two unrelated mESC datasets, log₁₀ space. Black line, identity. They are not very consistent (Pearson $R = 0.424$). **c**, Consistency of snRNA-seq and scRNA-seq estimates of burst size (Pearson $R = 0.918$) and splicing rate (Pearson $R = 0.972$) for 1,896 genes for which the bursty model fit well. Error bars, 99% confidence

regions determined by $\pm 2.576\sigma$. **d**, Uncertainties (that is, standard deviations of posterior in log₁₀ space) of snRNA-seq and scRNA-seq estimates of burst size and splicing rate. **e**, Consistency of scRNA-seq and snRNA-seq inferred 'degradation' (deg.) rates. Crimson line, best fit; black line, identity. The slope of the best fit line is significantly different from 1 (two-sided t -test; $(t, P) = (-11, 2.1 \times 10^{-27})$), and estimates based on snRNA-seq data tend to be slightly larger than estimates based on scRNA-seq data. **f**, Uncertainty after integrating snRNA-seq and scRNA-seq estimates versus lowest single-modality uncertainty across all genes. Note that all points are below the identity line, which reflects integration reducing uncertainty. **g**, Distribution of estimated uncertainty before and after integration. Horizontal lines, medians.

fluorescence data (for example, obtained via RNA sequential fluorescence in situ hybridization (seqFISH+)⁹⁴), and data reflecting information beyond RNA counts, such as protein abundances^{95,96} and chromatin accessibility^{97–99}. As different modalities may be better at capturing various aspects of the underlying biology (for example, snRNA-seq data are enriched for intronic reads¹³, and hence one might expect such data to be more informative about nuclear processes), integrating data across modalities allows different advantages to be simultaneously exploited.

Currently popular approaches to single-cell data integration are largely inspired by machine learning and typically use a shared latent space to make data from different modalities comparable^{100–104}. However, these approaches are not without problems; Tyler et al. have argued that they often remove biologically meaningful variation¹⁰⁵, and, more generally, it is unclear to what extent heuristic approaches to single-cell data analysis are trustworthy¹⁵.

We advocate for an alternative approach rooted in the mechanistic worldview. First, one defines a generative model whose biological component is shared but whose technical component differs across modalities; next, one fits the model to data; and, finally, one uses Bayes' rule to combine parameter estimates from different modalities according to their uncertainty (Fig. 5a). Monod supports this procedure by quantifying the uncertainty of its parameter estimates ('Rejection statistics and parameter uncertainty'). In the rest of this section, we walk through an example of this approach to integration.

Before integration, one must check whether estimates from different modalities are consistent, as inconsistencies may indicate model misspecification issues. We consider two example comparisons: between seqFISH+^{106,107} and 10x v3 scRNA-seq data¹⁰⁸, both collected from mouse embryonic stem cells (mESCs) in unrelated experiments; and between snRNA-seq and scRNA-seq data, both generated by 10x Genomics from the same mouse brain tissue sample^{109,110}. seqFISH-derived burst size estimates obtained using the procedure of Takei et al.¹⁰⁷ only coarsely agreed with Monod-derived estimates from analogous scRNA-seq data (Fig. 5b), possibly due to some combination of measuring slightly different features of the underlying biology and not adequately accounting for fluorescence-related technical noise. Curiously, the averages obtained from seqFISH and scRNA-seq agreed to an even lesser degree (Supplementary Fig. 9), suggesting that the burst sizes are, in a sense, more reliable than the raw expression values and that appropriately accounting for probe signal in the spirit of Xu et al.²⁹ may, in the future, help integrate the underlying biophysics. Meanwhile, Monod-derived estimates of burst size and splicing rate based on snRNA-seq and scRNA-seq mouse brain data were extremely consistent (Fig. 5c), which suggests that integration may be advisable in the latter case.

Interestingly, according to Monod's quantification of parameter uncertainty, estimates of burst size and splicing rate from snRNA-seq data are usually more precise than the corresponding scRNA-seq-based estimates (Fig. 5d; 84.9% of burst size estimates and 90.7% of splicing

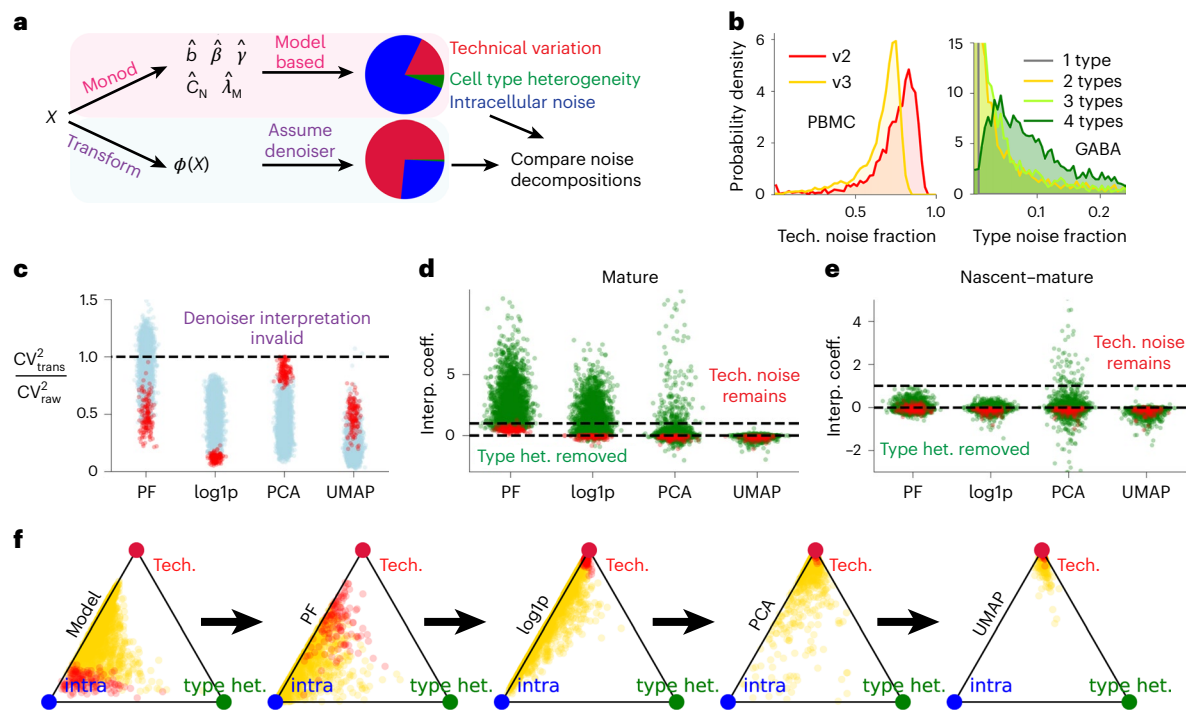


Fig. 6 | Monod enables quantitative assessment of common data transformations.

Monod can estimate how much observed variation is due to technical effects, cell type heterogeneity and intracellular noise; this noise decomposition can be used to assess whether common data transformations effectively erase biological signal. **a**, Schematic of the idea. A noise decomposition based on Monod-derived fits (in this figure, to the bursty model) can be compared to one that assumes a data transformation ϕ (for example, PF, log1p, PCA or UMAP) effectively removes technical noise. **b**, The Monod-derived noise decomposition behaves as expected. Technical (tech.) noise fractions derived from v2 PBMC data (red; 2,500 genes, three cell types) tend to be higher than those derived from matched v3 data (gold). Cell type heterogeneity fractions tend to be higher when more cell types are included, GABAergic neuron data (2,951 genes, four cell types). **c**, Applying many common transformations to scRNA-seq data tends to reduce the mature RNA CV² (64.1% for PF, 100% for log1p, 100% for PCA, 99.9% for UMAP), making a ‘denoiser’ interpretation reasonable. Red dots,

highly expressed (mature mean in 95th percentile) genes. This and the following panels depict glutamatergic neuron data (2,951 genes, six cell types). Trans, transformed. **d**, Post-transformation mature CV² tend to violate a model-agnostic upper bound b_{upper} or lower bound b_{lower} in an example preprocessing workflow that sequentially applies four data transformations (PF → log1p → PCA → UMAP). Mature CV² were written in the form $(1 - c)b_{lower} + cb_{upper}$, and values of c are plotted. $c > 1$ means that not all technical noise was removed, while $c < 0$ means that at least some cell type heterogeneity was removed. Coeff., coefficient; het., heterogeneity; interp., interpolation. **e**, Analogous plot for a measure of nascent–mature variation (‘Noise decomposition for assessing loss of signal after preprocessing’). **f**, Noise decomposition weights (mature RNA) throughout the example workflow are visualized as triangle plots. Applying more ‘denoising’ transformations misattributes more observed variation to technical effects (points concentrate in the top corner).

rate estimates are more precise), which is consistent with the intuition that snRNA-seq data ought to be more informative about nuclear processes. We obtained this result in spite of the rather smaller size of the single-nucleus dataset (Supplementary Table 8). On the other hand, on biophysical grounds, we do not expect snRNA-seq data to be informative about RNA degradation and instead expect mature RNA decay rate fits to measure nuclear export (‘Nuclear data integration’). Consistent with this intuition, we found that mature RNA decay rates were significantly different for each data type (Fig. 5e), with snRNA-seq-derived rates usually being somewhat higher.

Let μ_i denote the average parameter estimate given data from some modality, and let σ_i denote the corresponding uncertainty. Assuming that a sufficiently large number of cells are available, Bayes’ rule implies (‘Principled integration of single-cell technologies’) that the integrated estimate μ_{int} and its uncertainty σ_{int} satisfy

$$\frac{1}{\sigma_{int}^2} = \sum_j \frac{1}{\sigma_j^2}, \quad \mu_{int} = \sum_j \frac{\sigma_{int}^2}{\sigma_j^2} \mu_j,$$

that is, the uncertainty of the integrated estimate is smaller than each of the single-modality uncertainties. Applying the above formulas to estimates derived from snRNA-seq and scRNA-seq, we find that uncertainty is reduced, sometimes by up to 30% (Fig. 5f,g).

Assessing loss of biological signal after preprocessing

Because Monod fits mechanistic models to single-cell data, it enables a precise accounting of variation, in particular, variation due to (1) technical effects, (2) cell type heterogeneity and (3) intracellular noise. One useful way to summarize this accounting is in terms of a noise decomposition in the spirit of previous work^{48,111}, which expresses the coefficient of variation (CV) matrix associated with the data in terms of contributions due to each type of variation (‘Noise decomposition for assessing loss of signal after preprocessing’). For each gene and for three types of overall variation that one can track (nascent variance, mature variance and nascent–mature correlations), this decomposition allows one to generate a set of three weights that determine the ‘fraction’ of overall variation due to each source (Fig. 6a, top).

These noise fractions behave as one might intuitively expect. For example, 10x v2 human blood cell (peripheral blood mononuclear cell, or PBMC) data are generally noisier than matched 10x v3 data, and, accordingly, more of the overall variation is attributed to technical noise (Fig. 6b, left). When a GABAergic neuron dataset is analyzed with more cell types included (that is, with data from at least one of the four cell types left out), more variation is attributed to cell type heterogeneity (Fig. 6b, right).

Even if one is not especially interested in fitting mechanistic models to transcriptomic data, Monod’s accounting of variation allows it to play an interesting role: it can diagnose the extent to which potentially

biologically meaningful variation is lost or distorted after data are preprocessed. This function of Monod is important, as typical transcriptomic workflows involve a large number of heuristic correction steps⁶, including ‘cell size’ normalization, proportional fitting (PF), log transformation (log1p) and dimensionality reduction techniques like principal-component analysis (PCA) and uniform manifold approximation and projection (UMAP)¹¹². At least part of the motivation for performing these preprocessing steps is to ‘denoise’ the data (that is, remove or reduce variation due to technical effects) so that the input to downstream analyses is more biologically meaningful. Here, we use Monod and glutamatergic neuron data (2,951 genes, six cell types) to interrogate this motivation.

If we take the ‘denoising’ interpretation of preprocessing steps at face value, the CV matrix associated with the transformed data should reflect cell type to cell type and intracellular variation, but not technical noise (‘Noise decomposition for assessing loss of signal after preprocessing’), which yields an alternative noise decomposition (Fig. 6a, bottom). Interpreting transformations as denoisers appears to be largely reasonable, because standard transformations, applied individually rather than sequentially, reduce the CV for most genes (Fig. 6c).

On the other hand, CVs of transformations applied sequentially often appear to indicate a loss of biologically meaningful variation. Assuming that transformations remove some combination of technical and intracellular noise but not cell type heterogeneity, which is usually important for downstream analyses⁶, one can derive fairly model-free upper and lower bounds for the CV. Upper-bound violations indicate that not all technical noise was removed, while lower-bound violations indicate that at least some variation due to cell type heterogeneity was removed. In an example workflow in which four transformations (PF, log1p, PCA and UMAP) were sequentially applied to the glutamatergic neuron data, lower-bound violations were extremely common after the last two transformations (Fig. 6d). Similar statements can be made about correlations between nascent and mature RNA (Fig. 6e) and nascent CVs (not shown). Visualizing transformation-associated noise decompositions suggests that, after more preprocessing of the raw data, more observed variation is misattributed to technical effects (Fig. 6f). These transformations are heuristic, and the ‘correct’ way to analyze their implicit assumptions is far from clear. However, we obtain similarly severe violations of model-free bounds using a simpler, single-species noise baseline (‘Single-species model-free bounds’), suggesting that such common practices as, for example, using normalized data in lieu of raw data^{105,113}, benchmarking methods based on the PCA space¹⁴, or directly analyzing the UMAP embedding^{15,16} may all distort the signal of interest in the underlying data to an unacceptable degree.

Although this analysis focuses on the distributions of individual genes, it yields additional troubling implications for the analysis of gene–gene relationships. The discovery of gene regulatory networks (GRNs) and gene modules often relies on leveraging gene–gene correlation matrices (‘Nascent–mature correlations and their connections to GRN inference’). We often do not know about the activity and identifiability of these relationships a priori: even if regulatory motifs are present in the underlying biology, they are typically mediated by unobserved proteins. Interestingly, the collection of multimodal data provides a natural positive control: nascent and mature RNA are causally related, and their correlation is higher than typical gene–gene correlations (Supplementary Fig. 15). However, the iterative application of transformations substantially distorts them. Counterintuitively, the supposed removal of technical noise often decreases or changes the sign of nascent–mature correlations (Supplementary Fig. 16). As a consequence, their strong signal is ‘diluted’: for instance, of the genes with high nascent–mature correlations (in the top quartile of the mature–mature correlation distribution), the values stay in the top quartile only 60–80% of the time when a transformation is applied (Supplementary Figs. 17 and 18). For lower-magnitude nascent–mature

correlations, the preservation appears to be considerably worse, and transformations do not recover the biological signal.

The troubling conclusion is that these known, simple and biologically meaningful molecule–molecule relationships cease to be distinctly identifiable as the consequence of a standard preprocessing step for analysis. By contrast, the ‘underlying’ biological correlations implied by successful Monod fits are strictly higher than those observed in the raw data, matching intuition (Supplementary Fig. 19). Monod cannot yet model gene–gene relationships. However, there is existing work on biophysical GRN inference¹¹⁴, and we anticipate that these problems are amenable to a mechanistically satisfactory analysis by treating coupling through co-bursting modules¹¹⁵ or underlying chromatin states¹¹⁶.

Discussion

The Monod method we have proposed facilitates principled, mechanistic inference and treatment of multimodal scRNA-seq data and redefines common single-cell analysis tasks through biophysical data representation. Our work also demonstrates mechanistic inference at scale from relatively standard sequencing data types, which would otherwise entail more laborious and error-prone protocols to measure the relevant kinetics^{117,118}. While Monod is currently restricted to only a handful of transcriptional models that are tractable by quadrature, technical noise models comprising catalysis or dropout, the two modalities of nascent and mature RNA, correlation structure that does not include intergene relationships, and heterogeneity structures comprising a single homogeneous cell type at a time, many of these assumptions can be relaxed and extended. This is in contrast to current approaches to scRNA-seq analysis that correspond to much more simplistic, unrealistic, and opaque models of biology that make contradictory assumptions and violate fundamental constraints. Advancements in combining machine learning techniques with mechanistic inference^{55,119–121} demonstrate how the Monod framework could be extended in the future to more complex biophysical settings and models without analytical solutions. Likewise, the biophysical models in Monod provide a basis for integrating other single-cell measurements, such as chromatin information¹¹⁶ and protein counts^{37,120}.

We also note that all examples here have demonstrated inference across predefined, homogeneous populations of statistically equivalent cells; the design of Monod presupposes that it is applied downstream of cell type identification. However, there are potential advantages to applying this framework upstream: the standard procedure for clustering or cell type determination is based on mature mRNA counts only, and, without a model of the data in mind, it is not necessarily clear how to take advantage of nascent mRNA data³¹. This results in discarding a large fraction of reads containing valuable information and raises unsolved problems regarding the ‘correct’ way to, for example, co-cluster snRNA-seq and scRNA-seq data, which have substantially different RNA distributions.

As an alternative, we can use the biophysical basis of Monod to define the clusters of interest, that is, learn ‘clusters’ or ‘cell types’ of cells that display similar dynamics for the cellular processes of interest³¹. This amounts to fitting a mixture model representation of the model class we consider to learn which populations of cells emerge with shared bursting, splicing and degradation kinetics³¹. The same mechanistic foundation can be generalized and coupled to a continuous mixture representation to model and infer continuous cell states¹²² and trajectories¹²³.

Overall, even with simple systems such as the bursty transcription model for integrating nascent and mature RNA counts, we have demonstrated the potential for interesting discovery and generalization. Subtle differences and uncharacterized behaviors between cells in various states of development, treatment and disease can be identified and ascribed to biophysical phenomena; biophysical hypotheses can be compared for their ability, or lack thereof, to capture

important dataset features; data from multiple sequencing and imaging approaches can be described in a common framework; and the balance of technical noise and biological variation can be described and investigated in a self-consistent fashion. In light of the ever-growing complexity and resolution of high-throughput genomics data, Monod demonstrates a framework poised to take advantage of other multimodal data types, and to parametrize and generate more complex mechanistic hypotheses.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41592-025-02832-x>.

References

- Shekhar, K. et al. Comprehensive classification of retinal bipolar neurons by single-cell transcriptomics. *Cell* **166**, 1308–1323 (2016).
- Yao, Z. et al. A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain. *Nature* **624**, 317–332 (2023).
- Wagner, D. E. et al. Single-cell mapping of gene expression landscapes and lineage in the zebrafish embryo. *Science* **360**, 981–987 (2018).
- Regev, A. et al. Science forum: the human cell atlas. *eLife* **6**, e27041 (2017).
- Chari, T. et al. Whole-animal multiplexed single-cell RNA-seq reveals transcriptional shifts across *Clytia* medusa cell types. *Sci. Adv.* **7**, eabh1683 (2021).
- Luecken, M. D. & Theis, F. J. Current best practices in single cell RNA seq analysis: a tutorial. *Mol. Syst. Biol.* **15**, e8746 (2019).
- Kharchenko, P. V. The triumphs and limitations of computational methods for scRNA-seq. *Nat. Methods* **18**, 723–732 (2021).
- Heumos, L. et al. Best practices for single-cell analysis across modalities. *Nat. Rev. Genet.* **24**, 550–572 (2023).
- Kobak, D. & Berens, P. The art of using t-SNE for single-cell transcriptomics. *Nat. Commun.* **10**, 5416 (2019).
- Hu, Q. & Greene, C. S. Parameter tuning is a key part of dimensionality reduction via deep variational autoencoders for single cell RNA transcriptomics. *Pac. Symp. Biocomput.* **24**, 362–373 (2019).
- Raimundo, F., Vallot, C. & Vert, J.-P. Tuning parameters of dimensionality reduction methods for single-cell RNA-seq analysis. *Genome Biol.* **21**, 212 (2020).
- Cooley, S. M., Hamilton, T., Aragones, S. D., Ray, J. C. J. & Deeds, E. J. A novel metric reveals previously unrecognized distortion in dimensionality reduction of scRNA-seq data. Preprint at *bioRxiv* <https://doi.org/10.1101/689851> (2020).
- Chamberlin, J. T., Lee, Y., Marth, G. T. & Quinlan, A. R. Differences in molecular sampling and data processing explain variation among single-cell and single-nucleus RNA-seq experiments. *Genome Res.* **34**, 179–188 (2024).
- Ahlmann-Eltze, C. & Huber, W. Comparison of transformations for single-cell RNA-seq data. *Nat. Methods* **20**, 665–672 (2023).
- Chari, T. & Pachter, L. The specious art of single-cell genomics. *PLoS Comput. Biol.* **19**, e1011288 (2023).
- Gorin, G., Fang, M., Chari, T. & Pachter, L. RNA velocity unraveled. *PLoS Comput. Biol.* **18**, e1010492 (2022).
- Kiselev, V. Y., Andrews, T. S. & Hemberg, M. Challenges in unsupervised clustering of single-cell RNA-seq data. *Nat. Rev. Genet.* **20**, 273–282 (2019).
- Mimitou, E. P. et al. Scalable, multimodal profiling of chromatin accessibility, gene expression and protein levels in single cells. *Nat. Biotechnol.* **39**, 1246–1258 (2021).
- Elowitz, M. B., Levine, A. J., Siggia, E. D. & Swain, P. S. Stochastic gene expression in a single cell. *Science* **297**, 1183–1186 (2002).
- Raser, J. M. & O’Shea, E. K. Noise in gene expression: origins, consequences, and control. *Science* **309**, 2010–2013 (2005).
- Gillespie, D. T. Stochastic simulation of chemical kinetics. *Annu. Rev. Phys. Chem.* **58**, 35–55 (2007).
- Guillemin, A. & Stumpf, M. P. H. Noise and the molecular processes underlying cell fate decision-making. *Phys. Biol.* **18**, 011002 (2020).
- Gorin, G., Vastola, J. J., Fang, M. & Pachter, L. Interpretable and tractable models of transcriptional noise for the rational design of single-molecule quantification experiments. *Nat. Commun.* **13**, 7620 (2022).
- Gorin, G. & Pachter, L. Length biases in single-cell RNA sequencing of pre-mRNA. *Biophys. Rep.* **3**, 100097 (2023).
- Gorin, G., Vastola, J. J. & Pachter, L. Studying stochastic systems biology of the cell with single-cell genomics data. *Cell Syst.* **14**, 822–843 (2023).
- Gorin, G. & Pachter, L. New and notable: revisiting the ‘two cultures’ through extrinsic noise. *Biophys. J.* **123**, 1–3 (2024).
- Raj, A., Peskin, C. S., Tranchina, D., Vargas, D. Y. & Tyagi, S. Stochastic mRNA synthesis in mammalian cells. *PLoS Biol.* **4**, e309 (2006).
- Bokes, P., King, J. R., Wood, A. T. A. & Loose, M. Exact and approximate distributions of protein and mRNA levels in the low-copy regime of gene expression. *J. Math. Biol.* **64**, 829–854 (2012).
- Xu, H., Skinner, S. O., Sokac, A. M. & Golding, I. Stochastic kinetics of nascent RNA. *Phys. Rev. Lett.* **117**, 128101 (2016).
- Golding, I., Paulsson, J., Zawilski, S. M. & Cox, E. C. Real-time kinetics of gene activity in individual bacteria. *Cell* **123**, 1025–1036 (2005).
- Chari, T., Gorin, G. & Pachter, L. Biophysically interpretable inference of cell types from multimodal sequencing data. *Nat. Comput. Sci.* **4**, 677–689 (2024).
- Sullivan, D. K. et al. Accurate quantification of nascent and mature RNAs from single-cell and single-nucleus RNA-seq. *Nucleic Acids Res.* **53**, gkae1137 (2024).
- Sullivan, D. K. kallisto, bustools and kb-python for quantifying bulk, single-cell and single-nucleus RNA-seq. *Nat. Protoc.* **20**, 587–607 (2025).
- Jahnke, T. & Huisinga, W. Solving the chemical master equation for monomolecular reaction systems analytically. *J. Math. Biol.* **54**, 1–26 (2006).
- Vastola, J. J. Solving the chemical master equation for monomolecular reaction systems and beyond: a Doi–Peliti path integral view. *J. Math. Biol.* **83**, 48 (2021).
- Svensson, V. Droplet scRNA-seq is not zero-inflated. *Nat. Biotechnol.* **38**, 147–150 (2020).
- Singh, A. & Bokes, P. Consequences of mRNA transport on stochastic variability in protein levels. *Biophys. J.* **103**, 1087–1096 (2012).
- Dar, R. D. et al. Transcriptional burst frequency and burst size are equally modulated across the human genome. *Proc. Natl Acad. Sci. USA* **109**, 17454–17459 (2012).
- Halpern, K. B. et al. Bursty gene expression in the intact mammalian liver. *Mol. Cell* **58**, 147–156 (2015).
- Corrigan, A. M., Tunnacliffe, E., Cannon, D. & Chubb, J. R. A continuum model of transcriptional bursting. *eLife* **5**, e13051 (2016).
- Fukaya, T., Lim, B. & Levine, M. Enhancer control of transcriptional bursting. *Cell* **166**, 358–368 (2016).
- Nicolas, D., Phillips, N. E. & Naef, F. What shapes eukaryotic transcriptional bursting? *Mol. Biosyst.* **13**, 1280–1290 (2017).
- Rodriguez, J. & Larson, D. R. Transcription in living cells: molecular mechanisms of bursting. *Annu. Rev. Biochem.* **89**, 189–212 (2020).
- Ham, L., Brackston, R. D. & Stumpf, M. P. H. Extrinsic noise and heavy-tailed laws in gene expression. *Phys. Rev. Lett.* **124**, 108101 (2020).

45. Gorin, G., Yoshida, S. & Pachter, L. Assessing Markovian and delay models for single-nucleus RNA sequencing. *Bull. Math. Biol.* **85**, 114 (2023).
46. Leier, A. & Marquez-Lago, T. T. Delay chemical master equation: direct and closed-form solutions. *Proc. R. Soc. Math. Phys. Eng. Sci.* **471**, 20150049 (2015).
47. Desai, R. V. et al. A DNA repair pathway can regulate transcriptional noise to promote cell fate transitions. *Science* **373**, eabc6506 (2021).
48. Hilfinger, A. & Paulsson, J. Separating intrinsic from extrinsic fluctuations in dynamic biological systems. *Proc. Natl Acad. Sci. USA* **108**, 12167–12172 (2011).
49. Robinson, M. D., McCarthy, D. J. & Smyth, G. K. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* **26**, 139–140 (2010).
50. Anders, S. & Huber, W. Differential expression analysis for sequence count data. *Genome Biol.* **11**, R106 (2010).
51. Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* **15**, 550 (2014).
52. Hansen, M. M. K., Desai, R. V., Simpson, M. L. & Weinberger, L. S. Cytoplasmic amplification of transcriptional noise generates substantial cell-to-cell variability. *Cell Syst.* **7**, 384–397 (2018).
53. García-Blay, Ó. et al. Multimodal screen identifies noise-regulatory proteins. *Dev. Cell* **60**, 133–151 (2025).
54. Calia, G. P., Chen, X., Zuckerman, B. & Weinberger, L. S. Comparative analysis between single-cell RNA-seq and single-molecule RNA FISH indicates that the pyrimidine nucleobase idoxuridine (IdU) globally amplifies transcriptional noise. Preprint at *bioRxiv* <https://doi.org/10.1101/2023.03.14.532632> (2023).
55. Huang, Z. et al. Deep learning linking mechanistic models to single-cell transcriptomics data reveals transcriptional bursting in response to DNA damage. *eLife* <https://doi.org/10.7554/eLife.100623.2> (2024).
56. Yao, Z. et al. A transcriptomic and epigenomic cell atlas of the mouse primary motor cortex. *Nature* **598**, 103–110 (2021).
57. Mayère, C. et al. Single-cell transcriptomics reveal temporal dynamics of critical regulators of germ cell fate during mouse sex determination. *FASEB J.* **35**, e21452 (2021).
58. Stelzer, G. et al. The GeneCards suite: from gene data mining to disease genome sequence analyses. *Curr. Protoc. Bioinformatics* **54**, 1.30.1–1.30.33 (2016).
59. Lee, S. C.-W. & Abdel-Wahab, O. Therapeutic targeting of splicing in cancer. *Nat. Med.* **22**, 976–986 (2016).
60. Lu, H. et al. Single-cell map of dynamic cellular microenvironment of radiation-induced intestinal injury. *Commun. Biol.* **6**, 1248 (2023).
61. Hwang, W. L. et al. Single-nucleus and spatial transcriptome profiling of pancreatic cancer identifies multicellular dynamics associated with neoadjuvant treatment. *Nat. Genet.* **54**, 1178–1191 (2022).
62. Dyson, N. J. RB1: a prototype tumor suppressor and an enigma. *Genes Dev.* **30**, 1492–1502 (2016).
63. Touil, Y. et al. Colon cancer cells escape 5FU chemotherapy-induced cell death by entering stemness and quiescence associated with the c-Yes/YAP axis. *Clin. Cancer Res.* **20**, 837–846 (2014).
64. Huang, X. et al. Identification of genes related to 5-fluorouracil based chemotherapy for colorectal cancer. *Front. Immunol.* **13**, 887048 (2022).
65. Sahebnaasagh, R., Azizi, Z., Komeili-Movahhed, T., Zendejdel, K. & Ghahremani, M. H. In-silico and in-vitro investigation of key long non-coding RNAs involved in 5-fluorouracil resistance in colorectal cancer cells: analyses highlighting NEAT1 and MALAT1 as contributors. *Cureus* **16**, e66393 (2024).
66. Machkalyan, G., Hébert, T. E. & Miller, G. J. PIP5K1 suppresses etoposide-triggered apoptosis. *J. Mol. Signal.* **11**, 4 (2016).
67. Bhandari, K. & Ding, W.-Q. Protein arginine methyltransferases in pancreatic ductal adenocarcinoma: new molecular targets for therapy. *Int. J. Mol. Sci.* **25**, 3958 (2024).
68. Wang, X. et al. Characterization of LIMA1 and its emerging roles and potential therapeutic prospects in cancers. *Front. Oncol.* **13**, 1115943 (2023).
69. Hashemzahi, M. et al. Angiotensin receptor blocker losartan inhibits tumor growth of colorectal cancer. *EXCLI J.* **20**, 506–521 (2021).
70. Pothula, S. P. et al. Targeting HGF/c-MET axis in pancreatic cancer. *Int. J. Mol. Sci.* **21**, 9170 (2020).
71. Leclerc, G. J., Leclerc, G. M. & Barredo, J. C. Real-time RT-PCR analysis of mRNA decay: half-life of β -actin mRNA in human leukemia CCRF-CEM and Nalm-6 cell lines. *Cancer Cell Int.* **2**, 1 (2002).
72. Izdebska, M., Zielińska, W., Hałas-Wiśniewska, M. & Grzanka, A. Involvement of actin and actin-binding proteins in carcinogenesis. *Cells* **9**, 2245 (2020).
73. Arina, A. et al. Tumor-reprogrammed resident T cells resist radiation to control tumors. *Nat. Commun.* **10**, 3959 (2019).
74. Shadad, A. K., Sullivan, F. J., Martin, J. D. & Egan, L. J. Gastrointestinal radiation injury: symptoms, risk factors and mechanisms. *World J. Gastroenterol.* **19**, 185–198 (2013).
75. Augustin, R. C., Bao, R. & Luke, J. J. Targeting Cbl-b in cancer immunotherapy. *J. Immunother. Cancer* **11**, e006007 (2023).
76. Yu, M. et al. CD73 on cancer-associated fibroblasts enhanced by the A_{2B} -mediated feedforward circuit enforces an immune checkpoint. *Nat. Commun.* **11**, 515 (2020).
77. Qu, L. et al. NCAPD3 is a prognostic biomarker and is correlated with immune infiltrates in glioma. *Histol. Histopathol.* **39**, 1473–1484 (2024).
78. Zhong, Y. et al. Insulin-like growth factor 2 receptor is a key immune-related gene that is correlated with a poor prognosis in patients with triple-negative breast cancer: a bioinformatics analysis. *Front. Oncol.* **12**, 871786 (2022).
79. Xu, X. et al. Pan-cancer analysis of the role of MPP7 in human tumors. *Heliyon* **10**, e36148 (2024).
80. Dittmer, J. The biology of the Ets1 proto-oncogene. *Mol. Cancer* **2**, 29 (2003).
81. Phee, H., Mollenauer, M. N. & Weiss, A. Role of GIT2 in T cell migration and development (95.8). *J. Immunol.* **182**, 95.8 (2009).
82. O'Hagan, K. L., Miller, S. & Phee, H. Pak2 is essential for the function of Foxp3⁺ regulatory T cells through maintaining a suppressive T_{reg} phenotype. *Sci. Rep.* **7**, 17097 (2017).
83. Belenguer, G. et al. RNF43/ZNRF3 loss predisposes to hepatocellular-carcinoma by impairing liver regeneration and altering the liver lipid metabolic ground-state. *Nat. Commun.* **13**, 334 (2022).
84. Yue, F. et al. Loss of ZNRF3/RNF43 unleashes EGFR in cancer. *eLife* <https://doi.org/10.7554/eLife.95639.2> (2024).
85. Wei, J.-L. et al. GCH1 induces immunosuppression through metabolic reprogramming and IDO1 upregulation in triple-negative breast cancer. *J. Immunother. Cancer* **9**, e002383 (2021).
86. Zeng, Q. et al. LCP1 is a prognostic biomarker correlated with immune infiltrates in gastric cancer. *Cancer Biomark.* **30**, 105–125 (2021).
87. Pan, S. et al. Decreased expression of ARHGAP15 promotes the development of colorectal cancer through PTEN/AKT/FOXO1 axis. *Cell Death Dis.* **9**, 673 (2018).
88. Dou, B., Jiang, G., Peng, W. & Liu, C. OTULIN deficiency: focus on innate immune system impairment. *Front. Immunol.* **15**, 1371564 (2024).
89. Wang, W.-Y. et al. Interaction of FUS and HDAC1 regulates DNA damage response and repair in neurons. *Nat. Neurosci.* **16**, 1383–1391 (2013).

90. Martínez, D. et al. Discovery of BbX transcription factor in the Patagonian blennie: exploring expression changes following combined bacterial and thermal stress exposure. *Dev. Comp. Immunol.* **149**, 105056 (2023).
91. Kim, T. H., Zhou, X. & Chen, M. Demystifying ‘drop-outs’ in single-cell UMI data. *Genome Biol.* **21**, 196 (2020).
92. Ding, J. et al. Systematic comparison of single-cell and single-nucleus RNA-sequencing methods. *Nat. Biotechnol.* **38**, 737–746 (2020).
93. Habib, N. et al. Massively parallel single-nucleus RNA-seq with DroNc-seq. *Nat. Methods* **14**, 955–958 (2017).
94. Eng, C.-H. L. et al. Transcriptome-scale super-resolved imaging in tissues by RNA seqFISH+. *Nature* **568**, 235–239 (2019).
95. Bennett, H. M., Stephenson, W., Rose, C. M. & Darmanis, S. Single-cell proteomics enabled by next-generation sequencing or mass spectrometry. *Nat. Methods* **20**, 363–374 (2023).
96. Stoeckius, M. et al. Simultaneous epitope and transcriptome measurement in single cells. *Nat. Methods* **14**, 865–868 (2017).
97. Buenrostro, J. D., Wu, B., Chang, H. Y. & Greenleaf, W. J. ATAC-seq: a method for assaying chromatin accessibility genome-wide. *Curr. Protoc. Mol. Biol.* **109**, 21.29.1–21.29.9 (2015).
98. Buenrostro, J. D. et al. Single-cell chromatin accessibility reveals principles of regulatory variation. *Nature* **523**, 486–490 (2015).
99. Minnoye, L. et al. Chromatin accessibility profiling methods. *Nat. Rev. Methods Primers* **1**, 10 (2021).
100. Butler, A., Hoffman, P., Smibert, P., Papalexi, E. & Satija, R. Integrating single-cell transcriptomic data across different conditions, technologies, and species. *Nat. Biotechnol.* **36**, 411–420 (2018).
101. Stuart, T. et al. Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902 (2019).
102. Korsunsky, I. et al. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat. Methods* **16**, 1289–1296 (2019).
103. Ashuach, T., Reidenbach, D. A., Gayoso, A. & Yosef, N. PeakVI: a deep generative model for single-cell chromatin accessibility analysis. *Cell Rep. Methods* **2**, 100182 (2022).
104. Ashuach, T. et al. MultiVI: deep generative model for the integration of multimodal data. *Nat. Methods* **20**, 1222–1231 (2023).
105. Tyler, S. R., Guccione, E. & Schadt, E. E. Erasure of biologically meaningful signal by unsupervised scRNAseq batch-correction methods. Preprint at *bioRxiv* <https://doi.org/10.1101/2021.11.15.468733> (2023).
106. Shah, S. et al. Dynamics and spatial genomics of the nascent transcriptome by intron seqFISH. *Cell* **174**, 363–376 (2018).
107. Takei, Y. et al. High-resolution spatial multi-omics reveals cell-type specific nuclear compartments. Preprint at *bioRxiv* <https://doi.org/10.1101/2023.05.07.539762> (2023).
108. Khateb, M. et al. Transcriptomics, regulatory syntax, and enhancer identification in mesoderm-induced ESCs at single-cell resolution. *Cell Rep.* **40**, 111219 (2022).
109. 10x Genomics. *30k Mouse E18 Combined Cortex, Hippocampus and Subventricular Zone Cells Multiplexed, 12 CMOs, Brain 4* <https://www.10xgenomics.com/datasets/30-k-mouse-e-18-combined-cortex-hippocampus-and-subventricular-zone-cells-multiplexed-12-cm-os-3-1-standard-6-0-0> (2021).
110. 10x Genomics. *30k Mouse E18 Combined Cortex, Hippocampus and Subventricular Zone Nuclei Multiplexed, 12 CMOs, Brain Nuclei 4* <https://www.10xgenomics.com/datasets/30-k-mouse-e-18-combined-cortex-hippocampus-and-subventricular-zone-nuclei-multiplexed-12-cm-os-3-1-standard-6-0-0> (2021).
111. Swain, P. S., Elowitz, M. B. & Siggia, E. D. Intrinsic and extrinsic contributions to stochasticity in gene expression. *Proc. Natl Acad. Sci. USA* **99**, 12795–12800 (2002).
112. McInnes, L., Healy, J. & Melville, J. UMAP: uniform manifold approximation and projection for dimension reduction. Preprint at arxiv.org/abs/1802.03426 (2018).
113. Grima, R. & Esmenjaud, P.-M. Quantifying and correcting bias in transcriptional parameter inference from single-cell data. *Biophys. J.* **123**, 4–30 (2024).
114. Cao, Z. & Grima, R. Linear mapping approximation of gene regulatory networks with stochastic dynamics. *Nat. Commun.* **9**, 3305 (2018).
115. Gorin, G. & Pachter, L. Modeling bursty transcription and splicing with the chemical master equation. *Biophys. J.* **121**, 1056–1069 (2022).
116. Felce, C., Gorin, G. & Pachter, L. Biophysical model for joint analysis of chromatin and RNA sequencing data. *Phys. Rev. E* **110**, 064405 (2024).
117. Alpert, T., Herzog, L. & Neugebauer, K. M. Perfect timing: splicing and transcription rates in living cells. *Wiley Interdiscip. Rev. RNA* **8**, e1401 (2017).
118. Herzog, V. A. et al. Thiol-linked alkylation of RNA to assess expression dynamics. *Nat. Methods* **14**, 1198–1204 (2017).
119. Gorin, G., Carilli, M., Chari, T. & Pachter, L. Spectral neural approximations for models of transcriptional dynamics. *Biophys. J.* **123**, 2892–2901 (2024).
120. Sukys, A., Öcal, K. & Grima, R. Approximating solutions of the chemical master equation using neural networks. *iScience* **25**, 105010 (2022).
121. Cao, Z. et al. Efficient and scalable prediction of stochastic reaction–diffusion processes using graph neural networks. *Math. Biosci.* **375**, 109248 (2024).
122. Carilli, M. T., Gorin, G., Choi, Y., Chari, T. & Pachter, L. Biophysical modeling with variational autoencoders for bimodal, single-cell RNA sequencing data. *Nat. Methods* **21**, 1466–1469 (2024).
123. Fang, M., Gorin, G. & Pachter, L. Trajectory inference from single-cell genomics data with a process time model. *PLoS Comp. Biol.* **21**, e1012752 (2024).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2025

Methods

Overview of Monod inference

Here we briefly describe how models are defined and inference is performed in the Monod framework. Further details are available in the package API at <https://monod-examples.readthedocs.io>. The Monod package uses algorithms implemented in the NumPy¹²⁴, SciPy¹²⁵ and numdifftools¹²⁶ Python packages. Currently, all inference models require nascent and mature mRNA information, which can easily be obtained from realigning scRNA-seq datasets^{32,33}. For the datasets used in this study, unspliced and spliced (or nascent and mature mRNA) counts were obtained with kallisto|bustools (v.0.26.0)^{32,127}. We use the kb ref function with the --lamanno option to create an exon- and intron-containing reference and the kb count function with the --lamanno option, which takes in the FASTQ inputs to produce counts from the reads pseudo-aligned to the reference. For kallisto|bustools v.0.29.0, the --workflow nac option is used instead, and the mature and ambiguous counts are summed, which produces the spliced matrices of the --lamanno option. The details of processing for each specific dataset are reported below. See Supplementary Fig. 1 for a visual overview of the Monod inference procedure.

Biophysical model definition. The biological models in Monod, that is, the models describing the biophysical processes within the cell, all have the following generic structure:

$$\emptyset \xrightarrow{K} x_N \rightarrow x_M \rightarrow \emptyset. \quad (3)$$

For simplicity, x_N denotes nascent or unspliced mRNA species, whereas x_M denotes mature or spliced mRNA species. These concepts are not strictly synonymous; for example, ‘nascent’ species may refer to mRNA in the process of transcription. We therefore adopt this convention to emphasize that the models are generic, as long as a two-step process accurately describes the underlying physics (section 2.6 of Carilli et al.¹²²). K is a stochastic process governing mRNA arrivals, the transition from x_N to x_M is isomerization or splicing, and the removal of x_M is efflux or degradation. In this generic formulation, we do not specify the precise meaning of ‘ $\rightarrow$ ’, which may represent Markovian or deterministic transitions. To compute probabilities, we need to first evaluate the probability-generating function (PGF):

$$G(g_N, g_M) = \sum_{x_N, x_M=0}^{\infty} g_N^{x_N} g_M^{x_M} P(x_N, x_M), \quad (4)$$

where $P(x_N, x_M)$ is the joint probability mass function of the nascent and mature counts. We typically consider the log PGF in terms of shifted complex variables $u_z := g_z - 1$, where we introduce the generic transcript subscript $z \in \{N, M\}$.

To account for the imperfect sequencing process, we then define a set of technical noise models. All the models assume that reverse transcription, amplification and identification of the in vitro mRNA pool is fundamentally an independent and identically distributed process across all cells and molecules of a single gene, although the identically distributed assumption is relaxed across genes. For inference, we define the PGF H , which is the function composition of G with the PGFs $G_{z,t}$ of the sequencing processes:

$$H(u_N, u_M) = G(G_{N,t}(u_N) - 1, G_{M,t}(u_M) - 1). \quad (5)$$

Here t denotes the PGF for ‘technical’ sequencing processes. The governing rates of the biological and technical models constitute the inferred parameters θ . The full forms and derivations for each of the models implemented in Monod can be found in the Supplementary Methods^{128–133}.

Parameter inference procedure. To perform inference, Monod takes in the nascent and mature RNA molecule count matrices and a

user-selected biological and technical noise model. By default, Monod filters out genes with very low observed average and maximum expression ($\bar{x}_N \leq 0.01$, $\bar{x}_M \leq 0.01$, $\max x_N \leq 3$, $\max x_M \leq 3$) and those with excessively high observed maximum expression, which are too computationally intensive to fit ($\max x_N \geq 400$, $\max x_M \geq 400$). Genes are selected from the remaining genes for inference. The Monod inference procedure is then as follows:

1. Initialize a grid over possible values of technical sampling parameters (for the nascent and mature counts, respectively), which spans several orders of magnitude to capture physically realistic sampling rates of the transcriptome¹³⁴.
- For each grid point, in parallel:
 2. Initialize the biological parameters using the analytical moments of the desired model. These moments are derived and reported in the Supplementary Methods (section S1.4).
 3. Following the procedure outlined in refs. 28,37, evaluate the generating function, potentially using numerical quadrature and then invert it using a multivariate inverse fast Fourier transform to obtain the steady state probability distribution. This evaluation may be performed in parallel for each gene.
 4. Compute the KLD between the empirical histogram and the analytically derived likelihood and add it across genes to obtain the objective function.
 5. Repeat steps 3 and 4, iteratively updating the biological parameters to minimize the total KLD. We use the SciPy optimize.optimize function¹²⁵ for optimization.

Once the optimization has been completed for each grid point,

6. Determine the optimal set of technical parameters as the grid point with the lowest total KLD summed over all genes. This produces a final, optimal tuple of technical sampling parameters and the associated, inferred biophysical parameters.

Runtime across models per technical sampling parameter grid point ranged from tens of seconds to a little over a minute for almost 3,000 genes (Supplementary Table 9). To assess memory usage, we used a profiler that quantifies it in units of mebibytes (MiB), or 2^{20} bytes, which is convenient since computer memory is allocated in powers of 2. Memory usage for the inference procedure was similar across datasets of varying numbers of cells, with a slight increase for an increased number of genes (Supplementary Table 10). For example, memory usage for the T cell sample from mouse intestines⁶⁰ for sample S0, which includes 536 cells, with ten genes was 303.3 MiB and with 1,000 genes was 343.7 MiB. Similarly, for sample S26 with 798 cells, the maximum memory usage was 311.8 MiB for ten genes and 337.1 MiB for 1,000 genes. The inference procedure itself took little memory, with the main bottleneck being loading and storing input and inference result objects.

Rejection statistics and parameter uncertainty. Once we have obtained parameter estimates for each gene, there are several metrics in Monod that are used to reject genes that demonstrate poor fits and/or model misspecification. Genes are rejected if (1) the Hellinger distances between the empirical histograms and the model-predicted probabilities were greater than 0.05 and (2) the model was rejected by a χ^2 test with $P = 0.05$. We adjust the χ^2 threshold with Bonferroni correction for the number of tested genes. The χ^2 test was implemented using the function `scipy.stats.mstats.chisquare`¹²⁵. In addition, we exclude genes that are near the bounds of the search grid (within 1% of bounds, relative to the size of the search domain).

To compare biophysical hypotheses and fits for genes under different model specifications, the AIC is calculated per model (i) as follows:

$$\text{AIC}_i = 2k_i - 2 \log(\hat{\mathcal{L}}_i), \quad (6)$$

where k_i is the number of parameters in model i and L_i is the likelihood of the observed counts under model i at the optimal parameters.

Uncertainty estimates for the inferred parameters or standard error (σ) values are calculated from the square root of the diagonals of the inverse FIM and used for error bar calculation. The FIM is calculated as the Hessian matrix of the KLDs between the empirical histograms and the distributions induced by the final inferred parameters. Plots of the gene fits, KLDs, and gene rejection statistics are automatically generated for the user as shown in Supplementary Figs. 1–3.

Mechanistic differential expression

Statistical testing. At the conclusion of the procedure in the Supplementary Methods (section S1.1.1), we obtain a set of fits, parameter uncertainties and goodness-of-fit criteria under one or more models. We seek to move beyond averages and explain the differences between single-cell samples and cell types in terms of biophysical distribution parameters, in the spirit of Munsky et al.¹³⁵. In this section, we propose and apply an approach for the identification of cell type differences that would be difficult to detect using standard average-based procedures and demonstrate its performance using an experiment studying transcriptional noise amplification.

Given a set of Monod fits, we can identify DE genes at the level of kinetic parameters (section S4.6 of Gorin and Pachter²⁴). Briefly, we define DE genes as those genes displaying high-magnitude variation between conditions or cell populations, as contrasted with genes for which variation is consistent with Gaussian random noise²⁴.

In the $n = 1$ scenario, typical statistical tests cannot be used. To identify DE genes, we perform orthogonal distance regression between the parameter values of the two conditions being compared (for each parameter θ). We then fit a normal distribution to the model residuals across genes. This distribution has a mean (putatively, a low-magnitude offset due to uncertainty in fitting the technical noise parameters) and a variance (putatively, the variance of residuals under the null hypothesis). Next, we identify genes with outlier residuals (high-magnitude variation), defined to have a Z statistic that yields $P < 0.01$. Outlier genes are removed, and the regression and residual fitting procedures are repeated, with iterative removal of outlier genes, incrementally restricting the genes to those that appear to follow the null hypothesis. The final set of outlier genes constitutes the DE genes. This procedure is implemented in Monod and may be used for coarse pairwise comparisons, but we do not use it in this study.

In the $n > 1$ scenario, we can use typical statistical tests to compare the parameters. We apply the t -test without the assumption of equal variance and restrict the analysis to genes that were successfully fit (decreasing n for some genes). To avoid overrejecting based on the t -test, we first debias the parameter estimates by running the $n = 1$ orthogonal distance regression procedure on pairs of samples and then subtracting the mean of the inferred null normal distribution of residuals to heuristically account for uncertainty in the technical noise parameters and encode the intuition that the residuals should be zero centered.

In this study, we use the notation DE- θ to indicate that the parameter θ exhibited a Bonferroni-corrected P value less than 0.1 and mean $\log_2$ (FC) greater than 1 (between the two cell types or conditions). We also compute the mean $\log_2$ (fold difference) in mature RNA averages between the cell types and conditions to identify DE- θ genes with a magnitude less than 1. In other words, we can extract genes that have detectably large differences in biophysical parameters but do not, on average, exhibit large differences in mature RNA averages μ_M .

Signatures of frequency modulation. In the bursty model of transcription (Supplementary Methods (section S1.1.1)), we fit the rate parameters $\log_{10}(\beta)$ and $\log_{10}(\gamma)$, setting the burst frequency k to unity. This is formally equivalent to fitting $\log_{10}(\frac{\beta}{k})$ and $\log_{10}(\frac{\gamma}{k})$; although transient dynamics of the model depend on three independent parameters, the steady state behavior of the system is completely characterized by two

parameter combinations. Hence, one cannot uniquely identify all three parameters given a single steady state dataset.

The models we present are not natively adapted to detect changes in k : to unambiguously distinguish between modulation of upstream and downstream processes, time-resolved data are mandatory. However, if the magnitudes of changes in $\log_{10}(\frac{\beta}{k})$ and $\log_{10}(\frac{\gamma}{k})$ are approximately equal, frequency modulation is the more plausible scenario (for example, section S7.10.3 of Gorin and Pachter²⁴).

We propose that the modulation of k can be motivated by biological argument. β and γ , the rates of splicing and degradation, use a one-step, first-order, memoryless reaction as a highly simplified representation of a series of chemical transformations effected in tandem with a spliceosome or a ribonuclease complex, respectively. However, spliceosomes and ribonucleases are promiscuous, whereas transcription is highly regulated. Therefore, we hypothesize that targeted modulation of burst frequency upstream at the gene locus is more mechanistically plausible than the synchronized and targeted modulation of the downstream processes.

If we assume that β and γ are constant between conditions or cell types, we can compute an estimate of k modulation between population 1 and population 2:

$$\begin{aligned}\Delta \log_{10}\left(\frac{\beta}{k}\right) &= \log_{10}\left(\frac{\beta_2}{k_2}\right) - \log_{10}\left(\frac{\beta_1}{k_1}\right), \\ \Delta \log_{10}\left(\frac{\gamma}{k}\right) &= \log_{10}\left(\frac{\gamma_2}{k_2}\right) - \log_{10}\left(\frac{\gamma_1}{k_1}\right), \\ \Delta \log_{10}(k) &\approx -\Delta \log_{10}\left(\frac{\beta}{k}\right) = \log_{10}(k_2) - \log_{10}(k_1) \\ &\approx -\Delta \log_{10}\left(\frac{\gamma}{k}\right) = \log_{10}(k_2) - \log_{10}(k_1).\end{aligned}\quad (7)$$

Therefore, if the approximate equality $\Delta \log_{10}(\frac{\beta}{k}) \approx \Delta \log_{10}(\frac{\gamma}{k})$ holds, we can propose that $\Delta \log_{10}(k)$ has a similar magnitude but the opposite sign. We average the two to estimate the burst frequency modulation:

$$\Delta \log_{10}(k) \approx -\frac{1}{2} \left(\Delta \log_{10}\left(\frac{\beta}{k}\right) + \Delta \log_{10}\left(\frac{\gamma}{k}\right) \right). \quad (8)$$

Data processing: mouse neurons. To illustrate the approach, we compared the parameters for glutamatergic and GABAergic cell types from four mouse datasets (B08, C01, F08 and H12) generated by the Allen Institute for Brain Science^{56,136}. We previously performed the fits and identified genes that suggested substantial parameter modulation²⁴. Here, we revisit the fits and summarize the key findings. We report the t -test results from these analyses in Supplementary Table 1.

We identified DE- θ genes as described above for the parameter values between the GABAergic and glutamatergic cell types, for which the mean $\log_2$ (fold difference) in mature RNA averages between the cell types also had magnitude less than 1. We also omitted data points that were discarded by goodness-of-fit testing, as described in ‘Rejection statistics and parameter uncertainty’.

Next, we averaged the mean $\log_2$ (FCs) in β and γ to obtain an estimate of the $\log_2$ (FC) in burst frequency, as in equation (8). We plotted the resulting aggregated FCs in burst size and burst frequency against each other, highlighting genes that were DE- θ for some biological θ but not DE- μ_M .

We colored these genes by their effect on noise. It is elementary to show that, if the mean remains constant, a decrease in b compensated by an increase in k or equivalently a decrease in β/k and γ/k leads the joint distribution of nascent and mature RNA to become bivariate Poisson. For example, if a gene was found to exhibit significantly higher b , β or γ in GABAergic cells, we assigned it to the GABAergic set, as this suggests relative noise amplification in this cell type. This analysis is shown in Fig. 1d.

The structure of this plot bears further discussion, as it provides a convenient summary of useful statistical properties. The solid diagonal

line denotes the set of b and k combinations that yield a constant mean (all other parameters held equal). The dashed diagonal lines are offset by unity and show the range of parameters that give averages with less than twofold change in the mean. The dashed vertical and horizontal lines correspond to no change in b and k , respectively. Qualitatively, moving toward the top right corresponds to increasing the mean, moving toward the bottom left corresponds to decreasing the mean, moving toward the top left corresponds to decreasing the noise to the Poisson limit and moving to the bottom right corresponds to increasing the noise.

To demonstrate the qualitative impact of noise modulation, we visualized the distributions as well as the fits in both cell types, based on data from the B08 dataset. We selected the genes *Nin* and *Bach2*, which are associated with neuronal development, as discussed by Gorin and Pachter²⁴. In addition, we computed mature count averages for these genes in each cell type. This demonstration is given in Fig. 1d, right.

Data processing: mouse embryonic stem cells. To demonstrate the potential of this approach for detecting broad trends in transcriptional modulation without replicates, we considered the transcriptomes of mESCs with and without IdU perturbations. This dataset was generated by Desai et al.^{47,54} to investigate the effect of IdU incorporation on transcriptional bursting properties; the authors found that the perturbation appeared to increase the noise genome wide but did not affect averages.

Preprocessing. We used kallisto | bustools v.0.26.0 to preprocess data ('Overview of Monod inference'), using a prebuilt *Mus musculus* genome from <https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest> (mm10, 2020-A version) to construct the reference. We obtained the raw FASTQ files for the DMSO (control) and IdU datasets, both of which were generated using 10x v2 chemistry. From kb count (as described in 'Overview of Monod inference'), we obtained unspliced (intron-containing) and spliced (non-intron-containing) RNA count matrices^{32,137}.

Filtering. We filtered the dataset to remove 'low-quality' cells or empty droplets. First, we removed all barcodes that did not pass the default kallisto | bustools filter. Next, we removed all cell barcodes that were associated with fewer than 4×10^3 molecular barcodes, computed over all genes, corresponding to standard knee plot filtering procedures.

Next, we used Monod 0.2.6.0 to extract genes with moderate to high expression. We removed genes based on the Monod filter described in 'Parameter inference procedure'. This procedure produced a set of 4,373 genes that met the thresholds in both cell populations. We randomly selected 2,000 genes for further analysis, ensuring that the genes analyzed in the original report (*Nanog*, *Sox2*, *Pou5f1*, *Klf4*, *Wdr83*, *Stx7*, *Hif1an*, *Mtpap*, *Farsa*, *Wipi2* and *Snd1*) were included.

Inference and analysis of biophysical parameters. To fit a mechanistic model, we used Monod 0.2.6.0. We set up a 20×21 grid over the $\{\log_{10}(C_N), \log_{10}(\lambda_M)\}$ domain listed in Supplementary Table 7.

At each grid point, we iterated over the 2,000 genes, using gradient descent to identify the conditional maximum likelihood estimate (MLE) of $\{\log_{10}(b), \log_{10}(\beta), \log_{10}(\gamma)\}$, where the rates β and γ are defined in units of burst frequency k . We used the conditional method-of-moments estimate as the starting point and performed 15 steps of gradient descent. The procedure was parallelized over up to 80 processors (Intel Xeon Gold 6152, 2.10GHz). Runtimes varied between 16 and 17 min.

To identify the optimal sampling parameters, we identified the grid point with the lowest total KLD, computed over all genes. To ensure that we obtained the true optima under the bursty model, we performed four rounds of fixed-point iteration. First, we rejected a subset of genes if they were detected by the χ^2 test with $P = 0.01$ with

a Bonferroni correction and their Hellinger distance from the data distribution exceeded 0.05. Next, we recalculated the optimum based on the remaining data and repeated the procedure. This procedure did not change the optimum for any of the datasets. Furthermore, we investigated the stability of the optima under gene subsampling and found them to be stable and consistent.

The discovered optima were not consistent between datasets (orange points, Supplementary Fig. 6), and the likelihood landscapes were rugged and inconclusive (hatched region, Supplementary Fig. 6). This observation is in agreement with our previous analyses of 10x v2 datasets (for example, panel a of figures in section S7.6 of ref. 24): the older v2 technology does not appear to provide enough information to identify the technical noise parameters. Therefore, we somewhat arbitrarily used the grid point closest to $\log_{10}(C_N) = -6.5$, $\log_{10}(\lambda_M) = -1.2$, near the optimum discovered for a mouse neuron dataset in Fig. 3e of ref. 24. We analyzed the datasets under that set of parameters, recomputing the goodness-of-fit statistics accordingly.

We computed the mean $\log_2$ (FC) in burst size and burst frequency (equation (8)) and plotted them against each other using the conventions in Fig. 1d. We omitted data points that were discarded by goodness-of-fit testing. Finally, we identified all genes with $\log_2$ (FC) greater than 1.5 in b as well as k , which demonstrated significant noise amplification. To focus on genes with biologically interesting effects, we selected only those that had a mature RNA mean greater than unity in at least one of the conditions and reported them.

Data processing: mouse germ cells. To assess the use of biophysical inference of the development timeline, we used the germ cell scRNA-seq data collected by Mayère et al.⁵⁷ over several stages of mouse embryo development. In this study, germ cells were collected from male and female mouse embryos from the E11 stage to the E16 stage, with biological replicates for each stage and sex.

Preprocessing. We used the unspliced and spliced count matrices provided in the original study⁵⁷.

Filtering. As described above, we filtered out 'low-quality' cells or empty droplets, removing cell barcodes with fewer than 10^4 molecular barcodes, computed over all genes, corresponding to standard knee plot filtering procedures. We then used Monod 0.2.6.0 to extract 3,000 genes with moderate to high expression ('Parameter inference procedure') and included genes discussed in the study⁵⁷ for analysis. Each embryonic stage and biological replicate was treated separately, as samples were not necessarily sequenced together. We used samples and stages with at least 50 cells.

Inference and analysis of biophysical parameters. To fit a mechanistic model, we used Monod 0.2.6.0. We set up a 10×11 grid over the $\{\log_{10}(C_N), \log_{10}(\lambda_M)\}$ domain listed in Supplementary Table 7. Optimal technical parameters, as described above, were fit for each germ cell sample.

We then rejected a subset of genes if they were detected by the χ^2 test with $P = 0.01$ with a Bonferroni correction and their Hellinger distance from the data distribution exceeded 0.05 (that is, genes with poor fits to the bursty transcription and sequencing model). Next, we recalculated the optimum based on the remaining data and repeated the procedure.

To make all inferred parameters comparable, we also performed 'debiasing', subtracting the constant term from the orthogonal distance regression of the parameters of each stage versus the earliest (E11) stage for comparison, as discussed in 'Statistical testing'. We obtained parameter uncertainties by calculating approximate 99% confidence intervals, that is, 2.756σ , using the σ estimates from the FIM ('Rejection statistics and parameter uncertainty'), and display them as error bars in the figures.

Inferred parameters and trends were compared to previous analyses in the original study, which used read depth-normalized and log-transformed expression to analyze *Tbrg4* trends and unspliced/spliced ratios⁵⁷.

Data processing: PDAC tumor samples. Biophysical inference was performed on malignant cells from PDAC residual tumors in a patient treated with FOLFIRINOX, 5-FU and CRT, a patient treated with CRTI and three untreated patients. We only used 10x v3 snRNA-seq samples from the original study.

Preprocessing. Nascent and mature (unspliced and spliced) count matrices were generated using kallisto | bustools v.0.29.0 with the --workflow nac option. Mature and ambiguous counts were summed so as to be equivalent to the spliced matrices from the earlier --lamanno workflow.

Filtering. Cells were filtered for cell barcodes in the original study that denoted malignant cells⁶¹. We then used Monod 0.2.6.0 to extract 3,000 genes with moderate to high expression ('Parameter inference procedure') and included genes discussed in the study⁶¹ for analysis. Each patient sample was treated separately, as samples were not necessarily sequenced together. We used samples that were run with 10x v3 technology, with at least 50 annotated malignant cells.

Inference and analysis of biophysical parameters. Using Monod 0.2.6.0, we set up a 5×6 grid over the $\{\log_{10}(C_N), \log_{10}(\lambda_M)\}$ domain listed in Supplementary Table 7. Optimal technical parameters were determined for each patient sample of malignant cells.

Genes were then rejected if they were detected by the χ^2 test with $P = 0.01$ with a Bonferroni correction and their Hellinger distance from the data distribution exceeded 0.05 (that is, genes with poor fits to the bursty transcription and sequencing model). We then recalculated the optimum based on the remaining data and repeated the rejection procedure. In total, 1,030 genes were not rejected across all samples and used for all downstream analysis.

As with the germ cell data above, we also performed 'debiasing', subtracting the constant term from the orthogonal distance regression of the parameters of samples from each stage versus an untreated patient sample for comparison ('Statistical testing'). FCs between samples were then calculated for the corrected parameters of the patient treated with CRT and the patient treated with CRTI versus all three untreated samples (thus $n = 3$ FC 'replicates').

To identify genes for which transcription (burst size b or burst frequency k) was altered ($|\log_2(\text{FC})| > 1$) consistently across all three comparisons, we denoted genes for which β/k (splicing) and y/k (degradation) FCs were of the same sign and similar magnitude (with a ratio of 1.5 or less) as likely denoting burst frequency changes. Otherwise, for genes with consistent changes (across all three replicates) in splicing or degradation that did not show near-identical FC behavior, the splicing and degradation rate FCs were treated independently. To compare FCs in mature RNA counts between samples, we summed counts across cells to obtain pseudobulk expression values per sample and then normalized these values as in DESeq2 (ref. 51). Specifically, this is achieved by creating a reference sample by taking the geometric mean of mature gene counts per gene across all samples, dividing the value per gene per sample by the corresponding value per gene in the reference sample and then finding the median of these ratios as the size factor per sample. Samples were finally normalized by dividing by the size factor.

Data processing: irradiated T cells. We applied Monod to T cells from the jejunum of mouse intestines following RIII⁶⁰ to infer biophysical parameters across the time course of recovery. Authors of the original study aimed to characterize the transcriptomic response of the intestinal microenvironment to RIII by treating wild-type C57BL/6J male mice with 15 Gy of abdominal radiation. Organs from mice before radiation

treatment (day 0) and several days after treatment (days 1, 3, 7 and 14) were collected, and single cells were extracted and sequenced (10x Genomics v2 chemistry).

Preprocessing. The kallisto | bustools v.0.29.0 with the --workflow nac option was used to generate nascent and mature counts matrices, summing ambiguous and mature counts matrices (which is equivalent to the 'spliced' count matrix of the earlier --lamanno workflow).

Filtering and inference. Cells were first subset to those given annotations in the original dataset⁶⁰. Choosing to focus on the immune response to radiation treatment, we selected T cells as the target cell type and then subset to samples with more than 100 T cells per sample. This left 11 samples across four time points (samples S0, S22, S23 and S26 from day 0; S2 and S32 from day 1; S10, S8 and S9 from day 3; and S38 and S40 from day 14).

For these 11 samples, we fit the bursty model with Poisson sampling noise using Monod 0.2.6.0 as described in 'Parameter inference procedure' over a 10-by-11 grid of $\{\log_{10}(C_N), \log_{10}(\lambda_M)\}$ within the ranges listed in Supplementary Table 7. We fit the same 3,000 genes across all samples and then performed goodness-of-fit testing at a manually set technical sampling parameter grid point ($-6.78, -1.25$). We rejected genes using a χ^2 cutoff of $P = 0.01$ with a Bonferroni correction per sample and if their Hellinger distance from the empirical count distribution was greater than 0.05 in any sample, leaving 1,182 final genes for analysis.

Analysis. Parameters were 'debaised' to remove sample-specific effects by regressing all genes that passed QC per sample against S38 (as it had the most genes that were not rejected after goodness-of-fit testing) and subtracting the mean of the orthogonal distance regression, as described in 'Statistical testing'.

To find statistically significant differences in parameters, we performed, for each day (days 1, 3 and 14 after radiation) against day 0 (control day), a t -test to the 'debaised' $\log_{10}$ of inferred burst sizes, relative splicing rates and relative degradation rates. We used scipy.stats.ttest_ind with unequal variances¹²⁵. To comparably find statistically significant differences in mature mean expression values, we first added counts per cell to obtain 'pseudobulked' expression values per sample and then normalized these values as in DESeq2 (ref. 51). Specifically, this was achieved by creating a reference sample by taking the geometric mean of mature gene counts per gene across all samples, dividing the value per gene per sample by the corresponding value per gene in the reference sample, and then finding the median of these ratios as the size factor per sample. Samples were finally normalized by dividing by the size factor. We performed a t -test with unequal variance per day against day 0 on the $\log_{10}$ of these normalized mature counts. We corrected the raw P values per parameter using a Bonferroni correction for the number of tested genes (1,182) to find FDRs. We flagged as significant all parameters with an FDR of less than or equal to 0.05.

We further calculated the average FC between days 1, 3 and 14 and day 0 (separately per day) by averaging the raw values per day and taking the ratio against the average of the raw values on day 0.

To identify genes with similar behavior across time points, we formed vectors per gene with parameter and mature expression FCs against day 0 for all days and took the Euclidean distance against the FC vector for a given target gene. This enables discovery of genes with similar parameter changes across all days after radiation that further result in similar expression changes.

Biophysical model selection and rejection

To compare biophysical hypotheses across genes and cell types, we fit five available transcriptional models with Monod (models are defined in the Supplementary Methods (section S1.1)), using scRNA-seq data for the mouse brain sample B08, originally generated by the Allen Institute for Brain Science³⁶.

Preprocessing. We used kallisto | bustools 0.26.0 to preprocess data as described in ‘Overview of Monod inference’, using a prebuilt *M. musculus* genome from <https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest> (mm10, 2020-A version). We obtained the raw FASTQ files for the dataset B08 (refs. 56,136), which was generated using 10x v3 single-cell chemistry. Next, we used the kb count function as described in ‘Overview of Monod inference’ to produce unspliced (intron-containing) and spliced (non-intron-containing) RNA count matrices^{32,137}.

Filtering. We filtered the dataset to remove ‘low-quality’ cells or empty droplets. First, we removed all barcodes that did not pass the default kallisto | bustools filter. Next, we removed all cell barcodes that were associated with fewer than 10^4 molecular barcodes, computed over all genes, corresponding to standard knee plot filtering procedures.

Next, we split the dataset by cell type and subtype annotations⁵⁶. We extracted seven classes: six glutamatergic subtypes (L2/3 IT, L5 IT, L6 IT, L5/6 NP, L6 CT and L6b) and their union (glutamatergic). We omitted low-abundance cell types (L6 IT Car3 and L5 ET, with fewer than ten barcodes) from analysis and inclusion in the ‘glutamatergic’ category. We then used Monod 0.2.6.0 to extract genes with moderate to high expression given the default filter (‘Parameter inference procedure’).

Inference of biophysical parameters for five models. For each of the 2,951 genes that passed previously described Monod 0.2.6.0 expression thresholds (‘Parameter inference procedure’) and each of six glutamatergic cell subtypes (L2/3 IT, L5 IT, L6 IT, L5/6 NP, L6 CT and L6b), we fit five biological models using Monod 0.2.6.0. In particular, we used model arguments to fit biological models of bursty transcription (bursty), constitutive transcription (constitutive), constitutive transcription with extrinsic noise (extrinsic), bursty transcription with delayed degradation (delay) and bursty transcription with delayed splicing (delayed splicing). Models are described in the Supplementary Methods. For each model, we also included Poisson technical noise with length bias for nascent or unspliced molecules and Poisson noise without length bias for spliced molecules, setting up a 10×11 grid over the $\{\log_{10}(C_N), \log_{10}(\lambda_N)\}$ domain listed in Supplementary Table 7.

Optimal technical parameters per dataset and cell type were discovered by iterating over all genes at each grid point, identifying the conditional MLEs of the biological model’s parameters (using gradient descent with a maximum of 15 steps) and then identifying the grid point with the lowest KLD computed over all genes. Biological parameters were initialized with the conditional method-of-moments estimate. Two rounds of this fixed-point iteration were performed to increase confidence in the achieved optima.

Rejection criteria and AIC weights. As described in ‘Rejection statistics and parameter uncertainty’, gene fits under each model are rejected based on Hellinger distance and the χ^2 test P value between the empirical histograms and fit distributions. In addition, we rejected genes that were too near the parameter domain bounds. The AIC metric, described in ‘Rejection statistics and parameter uncertainty’, was also used to compare multiple model fits for a given gene. In particular, we use AIC (or Akaike) weights to make model comparisons, where:

$$\text{AIC}_i \text{ weight} = e^{(\text{AIC}_{\min} - \text{AIC}_i)/2}. \quad (9)$$

AIC weights were normalized to sum to one, corresponding to the interpretation of these values as model posteriors¹³⁸. Genes were assigned to the model with the largest normalized AIC weight.

Principled integration of single-cell technologies

In this section, we describe a fairly general theoretical view of integrating data from different modalities as well as the specific approaches we use to treat single-nucleus and seqFISH+ data.

Bayes’ rule and integration. Warm-up: single-modality case. To begin, consider the single-modality case: let $\mathbf{x}_i$ denote the vector of observed counts after sequencing cell i (of n_c total cells), and let $p(\mathbf{x}_i|\theta)$ denote the corresponding parameterized likelihood. By Bayes’ rule, we have

$$p(\theta|\{\mathbf{x}_i\}) \propto_{\theta} \prod_{i=1}^{n_c} p(\mathbf{x}_i|\theta), \quad (10)$$

assuming that each cell is independent and that we have a uniform prior on θ . Especially when n_c is large, the posterior is generally tight around the MLE θ^* ; in particular, we can expand the likelihood as

$$\begin{aligned} & \exp \left\{ \sum_i \log p(\mathbf{x}_i|\theta) \right\} \\ &= \exp \left\{ \sum_i \log p(\mathbf{x}_i|\theta^*) + \frac{1}{2} \sum_{a,b} (\theta_a - \theta_a^*) \frac{\partial^2 \log p(\mathbf{x}_i|\theta)}{\partial \theta_a \partial \theta_b} \bigg|_{\theta=\theta^*} (\theta_b - \theta_b^*) + \dots \right\}. \end{aligned}$$

Assuming that n_c is sufficiently large, by the central limit theorem, we have¹³⁹

$$n_c \left(\frac{1}{n_c} \sum_i \frac{\partial^2 \log p(\mathbf{x}_i|\theta)}{\partial \theta_a \partial \theta_b} \right) \bigg|_{\theta=\theta^*} \approx n_c \mathbb{E}_x \left(\frac{\partial^2 \log p(\mathbf{x}|\theta)}{\partial \theta_a \partial \theta_b} \right) \bigg|_{\theta=\theta^*} = -n_c \mathcal{J}_{ab}(\theta^*), \quad (11)$$

where we used the definition of the FIM in the last step. We now have

$$\exp \left\{ \sum_i \log p(\mathbf{x}_i|\theta) \right\} \approx \exp \left\{ \log \hat{L} - \frac{n_c}{2} (\theta - \theta^*)^T \mathcal{J}(\theta^*) (\theta - \theta^*) \right\},$$

where $\hat{L}$ denotes the likelihood at $\theta = \theta^*$. Hence, when n_c is sufficiently large,

$$p(\theta|\{\mathbf{x}_i\}) \approx \mathcal{N} \left(\theta; \theta^*, \frac{1}{n_c} \mathcal{J}(\theta^*)^{-1} \right). \quad (12)$$

Multimodality case. Assume instead that we have access to data from multiple modalities. Because the mathematics looks essentially the same regardless of the number of modalities we use, we will assume two here for simplicity. Let $\mathbf{x}_i^{(1)}$ denote the vector of observed counts after sequencing cell i (of $n_c^{(1)}$ total cells) via modality 1, and let $\mathbf{x}_j^{(2)}$ denote the analogous vector after sequencing cell j (of $n_c^{(2)}$ total cells) via modality 2. We assume here that data from different modalities are independent, as this is true in practice. By Bayes’ rule, if we once again assume a uniform prior, we have

$$p(\theta|\{\mathbf{x}_i^{(1)}, \mathbf{x}_j^{(2)}\}) \propto_{\theta} \left(\prod_{i=1}^{n_c^{(1)}} p(\mathbf{x}_i^{(1)}|\theta) \right) \left(\prod_{j=1}^{n_c^{(2)}} p(\mathbf{x}_j^{(2)}|\theta) \right). \quad (13)$$

Assuming that both $n_c^{(1)}$ and $n_c^{(2)}$ are sufficiently large, by the previous argument, we approximately have

$$p(\theta|\{\mathbf{x}_i^{(1)}, \mathbf{x}_j^{(2)}\}) \propto_{\theta} \mathcal{N} \left(\theta; \theta_1^*, \frac{1}{n_c^{(1)}} \mathcal{J}_1(\theta_1^*)^{-1} \right) \mathcal{N} \left(\theta; \theta_2^*, \frac{1}{n_c^{(2)}} \mathcal{J}_2(\theta_2^*)^{-1} \right), \quad (14)$$

where θ_i^* is the MLE obtained by using only data from modality i and $\mathcal{J}_i(\theta_i^*)$ is the corresponding FIM. The posterior associated with a product of normally distributed likelihoods is well studied in the context of Bayesian cue combination; the result is $p(\theta|\{\mathbf{x}_i^{(1)}, \mathbf{x}_j^{(2)}\}) = \mathcal{N}(\theta; \mu_{\text{int}}, \Sigma_{\text{int}})$, where

$$\Sigma_{\text{int}}^{-1} = n_c^{(1)} \mathcal{J}_1(\theta_1^*) + n_c^{(2)} \mathcal{J}_2(\theta_2^*) \quad \mu_{\text{int}} = \Sigma_{\text{int}} \left(n_c^{(1)} \mathcal{J}_1(\theta_1^*) \theta_1^* + n_c^{(2)} \mathcal{J}_2(\theta_2^*) \theta_2^* \right). \quad (15)$$

If we additionally assume that the off-diagonal terms of the FIM (which control correlations between estimates of different parameters) are negligible, the above expressions simplify somewhat. For any given parameter θ_k , we have

$$p(\theta_k | \{\mathbf{x}_i^{(1)}, \mathbf{x}_j^{(2)}\}) \propto_{\theta_k} \mathcal{N}(\theta_k; \theta_{1k}^*, \sigma_{1k}^2) \mathcal{N}(\theta_k; \theta_{2k}^*, \sigma_{2k}^2), \quad (16)$$

where $\sigma_{ik}^2 := 1/(n_c^{(i)} J_{i,kk}(\theta_i^*))$. This implies that $p(\theta_k | \{\mathbf{x}_i^{(1)}, \mathbf{x}_j^{(2)}\}) = \mathcal{N}(\theta_k; \mu_{\text{int},k}, \sigma_{\text{int},k}^2)$, where

$$\frac{1}{\sigma_{\text{int},k}^2} = \frac{1}{\sigma_{1k}^2} + \frac{1}{\sigma_{2k}^2} \quad \mu_{\text{int},k} = \frac{\sigma_{\text{int},k}^2}{\sigma_{1k}^2} \theta_{1k}^* + \frac{\sigma_{\text{int},k}^2}{\sigma_{2k}^2} \theta_{2k}^*. \quad (17)$$

We use the above equations to determine integrated estimates and the associated uncertainties in ‘Principled integration of multiple modalities’ and Fig. 5.

Caveat about uncertainty in technical noise parameters. In practice, technical noise parameters appear to be much harder to infer than biology-related parameters (for example, burst size). Partly due to this issue, Monod treats these two types of parameters differently in its optimization process and only quantifies uncertainty about biology-related parameters. Technical noise parameters are optimized by grid search (C_N by λ_M), and, for each grid point, the other parameters are optimized via gradient descent. At the highest-likelihood technical noise grid point and MLE, uncertainty in biology-related parameters is quantified by numerically computing the Hessian.

In the context of the preceding derivation, we essentially assume that uncertainty about technical noise parameters is not the dominant contributor to uncertainty in other parameters and so can be neglected. One justification for this assumption is that the technical noise parameters C_N and λ_M are assumed to be the same for each gene, and hence much more data are used to infer them than the biology-related parameters (which only use data from individual genes). From another perspective, this approach is consistent with estimating, for example, ‘size factors’ for cells and samples^{51,140} from raw data and then fitting models conditional on these estimates and eliding any associated uncertainty.

Nuclear data integration. We would like to coherently integrate scRNA-seq and snRNA-seq data. To do so, we need to specify the relationship between the two modalities. We can establish such a relationship from first principles by making assumptions regarding the underlying biophysical processes. For example, by proposing that nascent RNA is restricted to the nucleus, we can reasonably assume that nascent RNA dynamics should be identical between the two modalities. On the other hand, the mature RNA distributions and dynamics may have substantial differences, as nuclei are depleted in this species relative to the entire cell. In this section, we propose a possible foundation for the integration of these modalities.

To compare the distributions of single-cell and single-nucleus datasets and explain them using a mechanistic argument, we used mouse neuron datasets generated by 10x Genomics.

Mechanistic model definition. To describe stochastic dynamics and sampling in a single-cell dataset, we use the formulation given in ‘Biophysical model definition’ and outlined in more detail by Gorin and Pachter²⁴. To connect this model to nuclear data, we note that, formally, it can arise from the following model:

$$\emptyset \xrightarrow{k} B \times X_N \xrightarrow{\beta} X_{M,\text{nuc}} \xrightarrow{\gamma_e} X_{M,\text{cyt}} \xrightarrow{\gamma_c} \emptyset. \quad (18)$$

where $X_{M,\text{nuc}}$ and $X_{M,\text{cyt}}$ are nuclear and cytoplasmic mature RNA species, respectively. The rate γ_e describes the efflux of nuclear RNA, whereas the rate γ_c describes the degradation of cytoplasmic RNA.

In the limit $\gamma_c \ll \gamma_e$ or $\gamma_c \gg \gamma_e$, the model in ‘Biophysical model definition’ approximately holds for cytoplasmic data: if one of these stages is considerably longer lived, the two-stage processing of mature RNA can be effectively described by a one-stage model. In this case, γ can be interpreted as the lower rate. We typically assume that the first limit is most relevant, although orthogonal data suggest that the details are highly gene and tissue dependent¹⁴¹. We note that it is, in principle, straightforward¹⁴⁵ to implement a model that explicitly incorporates both parameters; however, for computational facility, we use the simpler reduced model and discard genes that fail to fit it. On the other hand, for nuclear data, the model holds for $\gamma = \gamma_e$.

Preprocessing. We used kallisto | bustools 0.26.0 to preprocess data. We downloaded a prebuilt *M. musculus* genome from <https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest> (mm10, 2020-A version). To build intronic and exonic references, we used the kb ref function with the --lamanno option. We obtained the raw FASTQ files for the ‘Brain 4’ and ‘Brain Nuclei 4’ datasets from two multiplexing experiments, both generated using 10x v3 chemistry. We selected these datasets because they had the highest average molecule counts per cell in both technologies. Next, we used the kb count function with the --lamanno option as well as the -x 10xv3 whitelist option to quantify the datasets, producing unspliced (intron-containing) and spliced (non-intron-containing) RNA count matrices^{32,137}.

Filtering. We filtered the dataset to remove ‘low-quality’ cells or empty droplets. First, we removed all barcodes that did not pass the default kallisto | bustools filter. Next, we removed all cell barcodes that were associated with fewer than T molecular barcodes ($T = 3 \times 10^3$ for scRNA-seq and 6×10^3 for snRNA-seq), computed over all genes, corresponding to standard knee plot filtering procedures. In addition, we removed cells with more than 10^5 barcodes, as they may reflect obscure technical noise sources unique to single-nucleus data.

Next, we used Monod 0.2.6.0 to extract genes with moderate to high expression (‘Parameter inference procedure’). This procedure produced a set of 5,690 genes that met the thresholds in all of the cell populations. We randomly selected 2,000 genes for further analysis.

Inference and analysis of biophysical parameters. To fit a mechanistic model, we used Monod 0.2.6.0. We set up a 20×21 grid over the $\{\log_{10}(C_N), \log_{10}(\lambda_M)\}$ domain listed in Supplementary Table 7.

At each grid point, we iterated over the 2,000 genes, using gradient descent to identify the conditional MLE of $\{\log_{10}(b), \log_{10}(\beta), \log_{10}(\gamma)\}$, where the rates β and γ are defined in units of burst frequency k . We used the conditional method-of-moments estimate as the starting point and performed 15 steps of gradient descent. The procedure was parallelized over up to 15 processors (Intel Xeon Gold 6152, 2.10 GHz). Runtimes varied between 2.2 h for the whole-cell dataset and 3.8 h for the nuclear dataset.

To identify the optimal sampling parameters, we identified the grid point with the lowest total KLD, computed over all genes. To ensure that we obtained the true optima under the bursty model, we performed four rounds of fixed-point iteration. First, we rejected a subset of genes if they were detected by the χ^2 test with $P = 0.01$ with Bonferroni correction and their Hellinger distance from the data distribution exceeded 0.05. Next, we recalculated the optimum based on the remaining data and repeated the procedure. This procedure did not change the optimum for any of the datasets. Furthermore, we investigated the stability of the optima under gene subsampling and found them to be stable and consistent.

The optimum discovered for the single-nucleus dataset demonstrated noticeably higher molecule observation probabilities (orange points, Supplementary Fig. 6). This observation was supported by basic observations of the dataset statistics: despite the depletion of cytoplasmic RNA, the single-nucleus dataset had as much mature RNA as the

single-cell dataset and approximately half an order of magnitude more nascent RNA (Supplementary Fig. 8, left). To illustrate these trends, we computed the offset from the ratio of the dataset-wide means. In addition, the single-nucleus dataset appeared to exhibit lower noise levels (Supplementary Fig. 8, right).

From physical considerations, the two independent experiments, performed using different technologies, may not necessarily have the same sampling parameters. However, as the samples were taken from the same tissue, they should have the same physics of transcription and splicing. Therefore, we somewhat arbitrarily assumed that the single-cell optimum was sufficiently accurate and chose a set of single-nucleus sampling parameters that provided the lowest squared errors for the $\log_{10}(b)$ and $\log_{10}(\beta)$ parameters. As shown by the blue points in Supplementary Fig. 6, the optimum thus discovered lay approximately half an order of magnitude above the optimum for the single-cell data and within the top 5th percentile for the sampling parameter likelihood landscape (hatched region). We analyzed the single-nucleus data under that set of parameters, recomputing the goodness-of-fit statistics accordingly.

To illustrate the differences between the datasets, we plotted the inferred parameters and the identity line. To quantify uncertainty in the parameters, we exploited the FIM as described in section S4.3.4 of Gorin and Pachter's study²⁴; we visualized the error bars, which represent the 99% confidence intervals for the biological parameters, conditional on the sampling parameter values. Finally, we applied a *t*-test, implemented with `scipy.stats.ttest_ind`¹²⁵, to the pairs of single-cell and single-nucleus parameter estimates. We omitted genes rejected by goodness-of-fit procedures from these computations and visualizations.

Fluorescence data integration. Recent studies have quantified the number of intron-containing RNA molecules and mRNA molecules for thousands of genes in single cells in a spatially resolved manner using a series of fluorescent probes designed to hybridize to either intron-containing RNA transcripts or spliced mRNA transcripts^{106,107}. These studies report burst sizes for genes by averaging over intron-containing fluorescent spot intensities. However, comparing single-cell fluorescent-reported burst sizes to scRNA-seq count data is not immediately possible: fitting scRNA-seq data for burst sizes using Monod allows principled comparison of these two single-cell technologies.

We fit the bursty model of transcription to 10x v3 scRNA-seq data from mESCs¹⁰⁸ using Monod to find burst sizes per gene and then compared the inferred burst sizes to those obtained using seqFISH+ on mESCs.

Preprocessing: seqFISH+ in mESCs. We downloaded the reported fluorescence intensity values associated with intron-containing and mRNA molecule spots in mESCs from Takei et al.¹⁰⁷ for two biological replicates. For the two replicates, we found burst sizes per gene as the authors reported by averaging intron-containing spot intensities over all cells for a given gene.

Preprocessing: scRNA-seq in mESCs. We obtained raw 10x v3 FASTQ files from Khateb et al.¹⁰⁸ using Sequence Read Archive run accession number [SRR18364193](https://www.ncbi.nlm.nih.gov/sra/SRR18364193). We then quantified the dataset using kallisto | bustools 0.26.0 with a prebuilt *M. musculus* genome from <https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest> (mm10, 2020-A version). Specifically, we called kb count with the `--lamanno` workflow option and the `-x 10xv3` whitelist option to produce unspliced (intron-containing) and spliced (non-intron-containing) RNA count matrices^{32,137}.

Filtering. To remove low-quality cells, we filtered the data by first discarding all barcodes that did not pass the default kallisto | bustools

filter. We then also removed cells with fewer than 700 total unique molecule identifiers, leaving 12,167 high-quality cells.

We then subset to genes with moderate to high expression using Monod 0.2.6.0 ('Parameter inference procedure'). From genes that passed these thresholds, we randomly selected 2,000 genes to fit.

Inference and analysis of biophysical parameters. We used Monod 0.2.6.0 to fit the bursty model of transcription (Supplementary Methods (section S1.1.1)) with Poisson technical noise (with the default length bias capture for nascent molecules). We set up a 20×21 grid over the $\{\log_{10}(C_N), \log_{10}(\lambda_M)\}$ domain listed in Supplementary Table 7.

At each grid point, we iterated over the 2,000 genes, using gradient descent to identify the conditional MLE of $\{\log_{10}(b), \log_{10}(\beta), \log_{10}(\gamma)\}$, where the rates β and γ are defined in units of burst frequency k . We used the conditional method-of-moments estimate as the starting point and a maximum of 20 steps of gradient descent.

We identified optimal sampling parameters by finding the grid point with the lowest total KLD between predicted fits and empirical distributions, computed over all genes. Finally, we applied model selection criteria to reject poor fits by enforcing that they pass a χ^2 test with $P = 0.01$ (corrected for 2,000 tested hypotheses with Bonferroni correction) and that their Hellinger distance from the data distribution exceeded 0.05.

Noise decomposition for assessing loss of signal after preprocessing

To characterize the performance of typical data processing and dimensionality reduction techniques, we need a meaningful baseline for 'good' performance. In this section, we derive a nontrivial decomposition of the CV matrix, the diagonal entries of which correspond to the squared CVs of nascent and mature RNA. Under the assumption that standard transformations act as 'denoisers', that is, that they remove technical noise but not biologically meaningful variation (for example, due to differences between cell types), we show that transformations imply a noise decomposition generally different from one obtained by fitting a mechanistic model; the extent to which the two differ can be used to assess a loss of biological signal. We also derive more generic bounds on the post-transformation CV matrix. For simplicity, we assume throughout that a pre-annotated dataset from a single tissue is available and view the results of our noise decomposition analysis as conditional on the assumption that the cell type annotations are sufficiently accurate.

Generative model and general noise decomposition derivation. Let C denote cell type, $\mathbf{z}$ denote the vector of true molecule counts and $\mathbf{x}$ denote the vector of observed molecule counts. Assume that the joint probability mass function has the form

$$p(\mathbf{x}, \mathbf{z}, C) = p(\mathbf{x}|\mathbf{z}, C) p(\mathbf{z}|C) p(C). \quad (19)$$

In words: assume that the distribution of intracellular molecule counts is generally cell type dependent and that technical noise can depend on both the true number of molecules and cell type.

The noise decomposition we exploit in the main text is a straightforward consequence of the law of total covariance. In particular, two separate applications of it produce the equations

$$\begin{aligned} \text{Cov}(\mathbf{x}|C) &= \mathbb{E}_{\mathbf{z}|C}(\text{Cov}(\mathbf{x}|\mathbf{z}, C)) + \text{Cov}_{\mathbf{z}|C}(\mathbb{E}(\mathbf{x}|\mathbf{z}, C)) \\ \text{Cov}(\mathbf{x}) &= \mathbb{E}_C(\text{Cov}(\mathbf{x}|C)) + \text{Cov}_C(\mathbb{E}(\mathbf{x}|C)). \end{aligned} \quad (20)$$

Combining these results, we obtain an interesting decomposition of the observed covariance matrix:

$$\text{Cov}(\mathbf{x}) = \mathbb{E}_{\mathbf{z}, C}(\text{Cov}(\mathbf{x}|\mathbf{z}, C)) + \mathbb{E}_C(\text{Cov}_{\mathbf{z}|C}(\mathbb{E}(\mathbf{x}|\mathbf{z}, C))) + \text{Cov}_C(\mathbb{E}(\mathbf{x}|C)). \quad (21)$$

Overall observed variation is due to some combination of (1) technical noise, (2) intracellular noise and (3) cell type heterogeneity; the above decomposition makes this statement quantitatively precise, because the first term can be identified with the contribution of technical noise, the second with the contribution of intracellular noise and the third with the contribution of cell type heterogeneity. If there is no variation of some kind, the relevant term in the above decomposition is zero.

This decomposition can be converted into a decomposition of the CV matrix by pre- and post-multiplying by the inverse of a diagonal matrix $\tilde{M}$ for which the entries are the components of the observed mean $\mathbb{E}(\mathbf{x})$. The observed CV matrix $V_{\text{raw}} := \tilde{M}^{-1} \text{Cov}(\mathbf{x}) \tilde{M}^{-1}$ equals

$$\tilde{M}^{-1} \mathbb{E}_{\mathbf{z}, C} (\text{Cov}(\mathbf{x}|\mathbf{z}, C)) \tilde{M}^{-1} + \tilde{M}^{-1} \mathbb{E}_C (\text{Cov}_{\mathbf{z}|C}(\mathbb{E}(\mathbf{x}|\mathbf{z}, C))) \tilde{M}^{-1} + \tilde{M}^{-1} \text{Cov}_C(\mathbb{E}(\mathbf{x}|C)) \tilde{M}^{-1} = V_{\text{tech}} + V_{\text{intra}} + V_{\text{type}}. \quad (22)$$

By definition, each term in the above decomposition is a positive semidefinite matrix, which allows us to interpret each term as capturing a ‘fraction’ between zero and one of the overall variation. In particular, if we assume that $\mathbf{x} = (x_N, x_M)^T$ is a two-dimensional vector that measures nascent and mature RNA counts, then variation in nascent and mature counts can be decomposed as

$$1 = \frac{\mathbb{E}_{\mathbf{z}, C}(\text{var}(x_N|\mathbf{z}, C))}{\text{var}(x_N)} + \frac{\mathbb{E}_C(\text{var}_{\mathbf{z}|C}(\mathbb{E}(x_N|\mathbf{z}, C)))}{\text{var}(x_N)} + \frac{\text{var}_C(\mathbb{E}(x_N|C))}{\text{var}(x_N)} \\ 1 = \frac{\mathbb{E}_{\mathbf{z}, C}(\text{var}(x_M|\mathbf{z}, C))}{\text{var}(x_M)} + \frac{\mathbb{E}_C(\text{var}_{\mathbf{z}|C}(\mathbb{E}(x_M|\mathbf{z}, C)))}{\text{var}(x_M)} + \frac{\text{var}_C(\mathbb{E}(x_M|C))}{\text{var}(x_M)}. \quad (23)$$

The above can be obtained by pre- and post-multiplying the (now assumed 2×2) matrix $\text{Cov}(\mathbf{x})$ by the unit vectors $(1, 0)^T$ and $(0, 1)^T$, respectively. An analogous decomposition of variation in nascent–mature correlations can be obtained by pre- and post-multiplying by $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$:

$$D := \frac{\text{var}(x_N)}{\mathbb{E}(x_N)^2} + 2 \frac{\text{Cov}(x_N, x_M)}{\mathbb{E}(x_N)\mathbb{E}(x_M)} + \frac{\text{var}(x_M)}{\mathbb{E}(x_M)^2} \\ 1 = \frac{\frac{1}{\mathbb{E}(x_N)^2} \mathbb{E}_{\mathbf{z}, C}(\text{var}(x_N|\mathbf{z}, C)) + 2 \frac{\mathbb{E}_{\mathbf{z}, C}(\text{Cov}(x_N, x_M|\mathbf{z}, C))}{\mathbb{E}(x_N)\mathbb{E}(x_M)} + \frac{1}{\mathbb{E}(x_M)^2} \mathbb{E}_{\mathbf{z}, C}(\text{var}(x_M|\mathbf{z}, C))}{D} \\ + \frac{\frac{\mathbb{E}_C(\text{var}_{\mathbf{z}|C}(\mathbb{E}(x_N|\mathbf{z}, C)))}{\mathbb{E}(x_N)^2} + 2 \frac{\mathbb{E}_C(\text{Cov}_{\mathbf{z}|C}(\mathbb{E}(x_N|\mathbf{z}, C), \mathbb{E}(x_M|\mathbf{z}, C)))}{\mathbb{E}(x_N)\mathbb{E}(x_M)} + \frac{\mathbb{E}_C(\text{var}_{\mathbf{z}|C}(\mathbb{E}(x_M|\mathbf{z}, C)))}{\mathbb{E}(x_M)^2}}{D} \\ + \frac{\frac{\text{var}_C(\mathbb{E}(x_N|C))}{\mathbb{E}(x_N)^2} + 2 \frac{1}{\mathbb{E}(x_N)\mathbb{E}(x_M)} \text{Cov}_C(\mathbb{E}(x_N|C), \mathbb{E}(x_M|C)) + \frac{\text{var}_C(\mathbb{E}(x_M|C))}{\mathbb{E}(x_M)^2}}{D}. \quad (24)$$

Specializing the noise decomposition. The noise decomposition we just derived is fairly general, and we would like to incorporate some of the assumptions used by currently supported Monod models. These assumptions are as follows:

- We assume that each molecular species is observable.
- We assume that each molecule is sequenced separately (in probability theory language: that the distribution $p(\mathbf{x}|\mathbf{z}, C)$ is infinitely divisible).
- We assume that the average effect of technical noise on the true molecule count $\mathbf{z}$ does not depend on cell type, that is, that $\mathbb{E}(\mathbf{x}|\mathbf{z}, C) = \mathbb{E}(\mathbf{x}|\mathbf{z})$.

The first assumption implies that $\mathbf{x}$ and $\mathbf{z}$ are vectors of the same length. The second assumption implies that

$$p(\mathbf{x}|\mathbf{z}, C) = \prod_{k=1}^D p(x_{k_i}|z_k, C) = \prod_{k=1}^D \sum_{x_{k_1} + \dots + x_{k_{z_k}} = x_k} p(x_{k_1}|1, C) \dots p(x_{k_{z_k}}|1, C), \quad (25)$$

where $p(x_{k_i}|1, C)$ is the same for all $i = 1, \dots, z_k$. Our first two assumptions together imply that

$$\mathbb{E}(\mathbf{x}|\mathbf{z}, C) = A_C \mathbf{z} \\ \text{Cov}(\mathbf{x}|\mathbf{z}) = S_C \text{diag}(\mathbf{z}), \quad (26)$$

where A_C and S_C are (generally cell type-specific) diagonal matrices for which the diagonal entries are positive, and $\text{diag}(\mathbf{z})$ is the diagonal matrix formed from the components of $\mathbf{z}$. These moments imply that our covariance decomposition becomes

$$\text{Cov}(\mathbf{x}) = \mathbb{E}_C(S_C \text{diag}(\mathbb{E}(\mathbf{z}|C))) + \mathbb{E}_C(A_C \text{Cov}_{\mathbf{z}|C}(\mathbf{z}) A_C) + \text{Cov}_C(A_C \mathbb{E}(\mathbf{z}|C)). \quad (27)$$

Our last assumption implies that $A_C = A$, that is, that it does not depend on cell type. We now have

$$\text{Cov}(\mathbf{x}) = \mathbb{E}_C(S_C \text{diag}(\mathbb{E}(\mathbf{z}|C))) + A \{\mathbb{E}_C(\text{Cov}_{\mathbf{z}|C}(\mathbf{z})) + \text{Cov}_C(\mathbb{E}(\mathbf{z}|C))\} A. \quad (28)$$

The equivalent CV matrix statement is that $V_{\text{raw}} := \tilde{M}^{-1} \text{Cov}(\mathbf{x}) \tilde{M}^{-1}$ equals

$$\tilde{M}^{-1} \mathbb{E}_C(S_C \text{diag}(\mathbb{E}(\mathbf{z}|C))) \tilde{M}^{-1} + M^{-1} \mathbb{E}_C(\text{Cov}_{\mathbf{z}|C}(\mathbf{z})) M^{-1} + M^{-1} \text{Cov}_C(\mathbb{E}(\mathbf{z}|C)) M^{-1} = V_{\text{tech}} + V_{\text{intra}} + V_{\text{type}} = V_{\text{tech}} + V_{\text{bio}}, \quad (29)$$

where M denotes the diagonal matrix for which the entries are the components of $\mathbb{E}(\mathbf{z})$. A notable feature of this decomposition is that the sum of the latter two terms is precisely the CV matrix associated with $\mathbf{z}$ rather than $\mathbf{x}$, that is, the analogous system without technical noise. Hence, there is a clean separation between ‘technical’ variation (the first term) and ‘biological’ variation (the last two terms).

If one assumes the catalysis technical noise model, as we do in the main text, $S_C = A$, and the expression slightly simplifies to

$$V_{\text{raw}} = \tilde{M}^{-1} + M^{-1} \mathbb{E}_C(\text{Cov}_{\mathbf{z}|C}(\mathbf{z})) M^{-1} + M^{-1} \text{Cov}_C(\mathbb{E}(\mathbf{z}|C)) M^{-1} \\ = V_{\text{tech}} + V_{\text{intra}} + V_{\text{type}} = V_{\text{tech}} + V_{\text{bio}}. \quad (30)$$

Equation (30) is the decomposition we exploit in the main text.

Measuring noise fractions. Here, we describe two approaches to estimating noise fractions given single-cell data. The first uses fits to a mechanistic model (for example, produced by Monod), while the second assumes that the role of preprocessing steps (for example, log1p) is to remove technical noise.

Model-based approach. Given fits to a (tractable) mechanistic model, one can use analytic expressions for the relevant means and covariances to determine each noise fraction. In ‘Assessing loss of biological signal after preprocessing’, we fit a model that assumes bursty transcription and catalysis technical noise; therefore, we will illustrate the approach in terms of the moments of that model.

Assuming bursty transcription, the relevant low-order moments of $p(\mathbf{z}|C)$ are

$$\mu_{N,C} = \frac{k_C b_C}{\beta_C} \quad \mu_{M,C} = \frac{k_C b_C}{\gamma_C} \\ \text{var}(z_N|C) = \mu_{N,C} (1 + b_C) \quad \text{Cov}(z_N, z_M|C) = \frac{k_C b_C^2}{\beta_C + \gamma_C} \\ \text{var}(z_M|C) = \mu_{M,C} \left(1 + \frac{b_C \beta_C}{\beta_C + \gamma_C} \right).$$

Because not all rates are identifiable from steady state data, Monod model fits assume that $k_C = 1$, and we do the same in what follows. The relevant low-order moments of $p(\mathbf{x}|\mathbf{z}, C)$ are

$$\mathbb{E}(x_N|\mathbf{z}, C) = \text{var}(x_N|\mathbf{z}, C) = C_N L z_N = \lambda_N z_N$$

$$\mathbb{E}(x_M|\mathbf{z}, C) = \text{var}(x_M|\mathbf{z}, C) = \lambda_M z_M,$$

where L denotes gene length and where we define $\lambda_N := C_N L$ to ease notation. We can now write the noise decomposition in terms of model parameters, as we have expressions for all the relevant moments. For example, the nascent squared CV and noise fractions are

$$\begin{aligned} V_{\text{raw},N} &= \frac{1}{\mu_N^2} \left\{ \frac{\mu_N}{\lambda_N} + \mathbb{E}_C(\mu_{N,C}(1+b_C)) + \text{var}_C(\mu_{N,C}) \right\} \\ f_{N,\text{tech}} &= \frac{\mu_N/\lambda_N}{\frac{\mu_N}{\lambda_N} + \mathbb{E}_C(\mu_{N,C}(1+b_C)) + \text{var}_C(\mu_{N,C})} \\ f_{N,\text{intra}} &= \frac{\mathbb{E}_C(\mu_{N,C}(1+b_C))}{\frac{\mu_N}{\lambda_N} + \mathbb{E}_C(\mu_{N,C}(1+b_C)) + \text{var}_C(\mu_{N,C})} \\ f_{N,\text{type}} &= \frac{\text{var}_C(\mu_{N,C})}{\frac{\mu_N}{\lambda_N} + \mathbb{E}_C(\mu_{N,C}(1+b_C)) + \text{var}_C(\mu_{N,C})}, \end{aligned} \quad (31)$$

where μ_N denotes $\mathbb{E}(z_N)$.

To compute the expectations over cell types, we use the observed cell type ratios and assume that they are sufficiently close to the ‘true’ ratios. From a ‘forward’ model construction perspective, we suppose that the sequencing process uniformly samples cells from the underlying population. Naturally, this may not be the case, and physiological considerations may dictate otherwise: for instance, some cells may not be amenable to sequencing due to morphology incompatible with dissociation and capture¹⁴². In other words, the true cell population distribution may well be unknowable, barring, for example, carefully calibrated orthogonal bulk RNA-seq experiments¹⁴³. From the considerably more fruitful ‘reverse’ data analytic perspective, this condition is guaranteed by construction: we define the entries of the categorical distribution based on abundances observed in the data and then, still considering the same cells and no others, analyze variability in the underlying biology.

Denoiser-based approach. A conceptually separate approach to estimating noise fractions does not involve model fits but raw data and preprocessed and/or transformed (for example, by applying a log1p transformation) data. This approach takes seriously the ‘denoiser’ interpretation of transformations, which is both part of the motivation for applying these transformations in the first place and appears empirically reasonable (Fig. 6c).

Let ϕ denote a transformation applied to observed RNA counts $\mathbf{x} := (x_N, x_M)^T$. If we assume that the only effect of transformations is to remove technical noise and not any biologically meaningful variation, we can identify $\phi(\mathbf{x})$ with the latent count vector $\mathbf{z} := (z_N, z_M)^T$. In the previous subsection, we found that (under certain assumptions) the observed CV matrix is equal to the CV matrix of the analogous system without technical noise, plus a contribution due to technical noise. Under the denoiser assumption, we then have

$$V_{\text{raw}} = V_{\text{tech}} + V_{\text{trans}}. \quad (32)$$

As both V_{raw} and V_{trans} can be measured, V_{tech} can be inferred from their difference. We can disentangle intracellular and type-to-type variation by applying the transformation separately to data from each cell type. By the law of total covariance and the denoiser assumption (equation (30)),

$$\begin{aligned} V_{\text{intra}} &:= M_{\text{trans}}^{-1} \mathbb{E}_C(\text{Cov}(\phi(\mathbf{x})|C)) M_{\text{trans}}^{-1} \\ V_{\text{type}} &:= M_{\text{trans}}^{-1} \text{Cov}_C(\mathbb{E}(\phi(\mathbf{x})|C)) M_{\text{trans}}^{-1}, \end{aligned} \quad (33)$$

where M_{trans} denotes the diagonal matrix formed from the transformed mean $\mathbb{E}(\phi(\mathbf{x}))$.

Transformations used. We considered four commonly used transformations: PF normalization (ϕ_1), log1p (ϕ_2), PCA (ϕ_3) and UMAP (ϕ_4)¹¹². Let

c denote the number of cells and g the number of genes. Let $\mathcal{X}_N \in \mathbb{N}^{c \times g}$ and $\mathcal{X}_M \in \mathbb{N}^{c \times g}$ denote the raw nascent and mature count matrices, respectively. Let $\mathcal{Y}_N$ and $\mathcal{Y}_M$ denote possibly transformed matrices of the same dimensions.

PF involves computing a correction factor

$$C_i := \frac{1}{c} \frac{\sum_j \mathcal{X}_{N,ij} + \mathcal{X}_{M,ij}}{\sum_j \mathcal{X}_{N,ij} + \mathcal{X}_{M,ij}}. \quad (34)$$

This factor involves averages over both cells and genes. The corresponding transformation is

$$\phi_1(\mathcal{X}_N)_{ij} = C_i \mathcal{X}_{N,ij} \quad \phi_1(\mathcal{X}_M)_{ij} = C_i \mathcal{X}_{M,ij}. \quad (35)$$

Log1p is simply defined by

$$\phi_2(\mathcal{Y}_N)_{ij} = \log(1 + \mathcal{Y}_{N,ij}) \quad \phi_2(\mathcal{Y}_M)_{ij} = \log(1 + \mathcal{Y}_{M,ij}). \quad (36)$$

The reconstructed PCA projection ϕ_3 is given by $\psi_3^{-1} \circ \psi_3$, where ψ_3 is implemented through the scikit-learn 1.0.1 function transform and ψ_3^{-1} is implemented through the function inverse_transform, both associated with a sklearn.decomposition.PCA object¹⁴⁴. The inverse transform is applied to ensure that the count matrix has the same dimensions as the original and hence to make the comparison more fair. We used 50 components to compute the PCA projection.

The reconstructed UMAP projection ϕ_4 is given by $\psi_4^{-1} \circ \psi_4$, where ψ_4 is implemented through the umap-learn 0.5.1 function transform and ψ_4^{-1} is implemented through the function inverse_transform, both associated with a umap.UMAP object¹¹². We used the 50-dimensional PCA projection and default parameters to compute the UMAP projection.

In Fig. 6c, PCA is applied to nascent and mature count matrices (without any additional transformations) separately. UMAP uses the same PCA decomposition in this panel. In the rest of Fig. 6, PCA directions are determined by finding the principal vectors associated with the sum of the nascent and mature count matrices.

In Fig. 6c, all transformations are applied individually. In the remaining panels, transformations are applied sequentially (PF $\rightarrow$ log1p $\rightarrow$ PCA $\rightarrow$ UMAP; $\phi_4 \circ \phi_3 \circ \phi_2 \circ \phi_1$), as they would be in a real pre-processing workflow; for example, in Fig. 6d, the label ‘PCA’ depicts results for data transformed first by PF and then by log1p and finally by PCA.

Model-free bounds. The assumption we use to associate transformations with noise decompositions is that transformations only remove technical noise; this assumption might be overly strict, as, for the purpose of many downstream analyses (for example, identifying cell types or subtypes by clustering), preserving intracellular variation may not be viewed as crucial. Here, we relax that assumption and assume instead that transformations remove all technical noise (as the purpose of applying transformations is generally to produce ‘cleaner’ data) and possibly some intracellular variation. This relaxed assumption implies (relatively) model-free upper and lower bounds for the CV matrix.

The upper bound corresponds to assuming that only technical noise is removed, as we did previously. Assuming the catalysis technical noise model (equation (30)), define

$$b_{\text{upper}} := \tilde{M}^{-1}. \quad (37)$$

Then $b_{\text{upper}} - V_{\text{trans}}$ is positive semidefinite. The lower bound corresponds to assuming that both technical and intracellular variation are completely removed, leaving only type-to-type variation. Motivated by equation (22), define

$$b_{\text{lower}} := \tilde{M}^{-1} \text{Cov}_C(\mathbb{E}(\mathbf{x}|C)) \tilde{M}^{-1}. \quad (38)$$

Then $V_{\text{trans}} - b_{\text{lower}}$ is positive semidefinite. Together, these results imply that

$$\begin{aligned} b_{\text{lower,NN}} &\leq V_{\text{trans,NN}} \leq b_{\text{upper,NN}} \\ b_{\text{lower,MM}} &\leq V_{\text{trans,MM}} \leq b_{\text{upper,MM}} \\ \mathbf{e}_{\text{II}}^T b_{\text{lower}} \mathbf{e}_{\text{II}} &\leq \mathbf{e}_{\text{II}}^T V_{\text{trans}} \mathbf{e}_{\text{II}} \leq \mathbf{e}_{\text{II}}^T b_{\text{upper}} \mathbf{e}_{\text{II}}, \end{aligned} \quad (39)$$

where we define $\mathbf{e}_{\text{II}} := (1/\sqrt{2}, 1/\sqrt{2})^T$. These are the ‘model-free’ bounds used in ‘Assessing loss of biological signal after preprocessing’. We use the second inequality (which bounds post-transformation mature RNA variation) in Fig. 6d and the last inequality (which bounds post-transformation nascent–mature correlations) in Fig. 6e.

Single-species model-free bounds. Interestingly, the derivation in ‘Model-free bounds’ can be made more generic yet. First, we relax the assumption of the technical noise model (equation (37)), essentially choosing to make no statements regarding potential retention of technical variability. In this scenario, we obtain a single lower bound per gene, the scalar version of equation (38):

$$b_{\text{lower}} = \frac{\text{var}_C(\mathbb{E}(x_M|C))}{M_M^2}, \quad (40)$$

that is, precisely the empirical, rather than theoretical, version of $f_{M,\text{type}}$ implied by equation (31) and conceptually a consequence of the law of total variance.

Next, we redefine the transformations in ‘Model-free bounds’ to only involve x_M , more closely matching the transformations actually used in practice but choosing to make no statements regarding the retention of covariance between nascent and mature species. This amounts to defining an analog to equation (34) as

$$C_i := \frac{1}{c} \frac{\sum_{i,j} x_{M,ij}}{\sum_j x_{M,ij}}, \quad (41)$$

and otherwise applying the standard transformations sequentially. In this scenario, we obtain a theoretical baseline for mature RNA (Supplementary Fig. 10a), analogous to the one derived in Model-free bounds. Given this baseline, we can inspect the effect of transformations and dimensionality reductions on the CV (Supplementary Fig. 10c), particularly on high-mean, high-variance genes (Supplementary Fig. 10b). As with the more generic case in equation (32), the denoiser interpretation induces ‘attributions’ of noise, with higher-expression genes losing a higher fraction of their CV (Supplementary Fig. 10d–g). With these results and the baseline, we can readily plot the fractions of remaining noise, as in Fig. 6d, but on a common scale rather than relative to the window between lower and upper bounds (Supplementary Fig. 10h–k). In addition, we computed the fraction of genes violating the bound at each step: 0%, 7.5%, 43.6% and 7.6% after PF, log1p, PCA and UMAP, respectively.

As the transformations are applied cumulatively, the distribution at step i may fall within the admissible region (above the theoretical baseline) but still be quantitatively degraded because it fell outside it at some step $i' < i$. To characterize the loss of quantitative information about cell type relationships, we plotted the same data points as above and colored them according to the history of the analysis. If a gene had ever violated the lower bound, we plotted it in a violet color. This analysis is shown in Supplementary Fig. 12. In addition, we computed the fraction of genes identified after each step. We found that 0%, 7.5%, 43.6% and 44.4% of the genes had at some point violated the bound after PF, log1p, PCA and UMAP, respectively.

Fig. 6f suggests that the Monod model and the normalizations effectively produce opposite attributions: when the model suggests that variability in high-expression genes (in red) is largely due to

biological effects, the normalizations disproportionately remove variance from those very genes, implicitly encoding that the variability in high-expression genes is largely due to dispensable technical effects. We can use the model-based approach in ‘Measuring noise fractions’ to estimate the biological variability attributions and agreement with the theoretical baseline (Supplementary Fig. 11), with attributions that fundamentally do not cohere with the normalization effects (Supplementary Fig. 13). Interestingly, this attribution, which nominally adds the $f_{M,\text{intra}}$ and $f_{M,\text{type}}$ attributions from the mature RNA analog of equation (31), mostly represents the former term: conflating all the cell subtypes together and running Monod once on the entire glutamatergic dataset produces practically identical results (Supplementary Fig. 14).

Nascent–mature correlations and their connections to GRN inference. The procedure in Model-free bounds is separately applied to nascent and mature count matrices, whereas the one in ‘Single-species model-free bounds’ is applied to the mature count matrix only to emphasize the consequences for the exonic (protein-coding) transcriptome the field is traditionally interested in. The same sequence of transformations may, in principle, be applied to the matrix $z := z^N \oplus z^M$, the concatenation of nascent and mature counts, which yields a $c \times 2g$ matrix³¹. Although the joint distribution of nascent and mature RNA can provide biological^{25,31,145} and technical^{146,147} insights somewhat different from either of the individual modalities or their sum, this approach is not typical. However, it is instructive in the unexpected context of GRNs.

GRN inference encompasses the identification of regulatory interactions between genes. The robust inference of such interactions is a key challenge in transcriptomics^{6,148–150}, with a broad variety of strategies and prospects^{148,149,151}. Although this subfield is far too broad to summarize in detail here, one key barrier is, conceptually, establishing causation from correlation: the strength of association between two genes may represent real regulatory motifs or merely reflect confounding. A related and more fundamental challenge is the biochemistry of regulation: sequencing reads out noise-corrupted proxies for RNA content, whereas regulation proceeds through protein and DNA interactions. In other words, even ostensibly known gene–gene interactions, if they are active in a particular system, may not be identifiable from RNA data alone.

The relationship between nascent and mature RNA provides a counterpoint. Mature RNA is produced from nascent RNA; causality is guaranteed. Although the process may have unobservable intermediate species, for example, transient RNA molecules with no polyA content, the proximity of the source and product is far closer than in protein-mediated processes. Finally, this process is guaranteed to occur, whereas some GRN links may be inhibited in a particular biological system.

As expected, the nascent–mature correlation distribution skews higher than the overall distribution of mature–mature correlations (Supplementary Fig. 15a). For a sizeable fraction of genes, the correlation between nascent and mature molecules for gene j is higher than intergene correlations between gene j and other genes $j' \neq j$. We quantify this by calculating the quantile of the nascent–mature correlation with respect to the empirical distribution of mature–mature correlations for each gene (Supplementary Fig. 15b).

GRN inference typically uses normalized data as an input^{148,150}. We may naturally hope that this preprocessing treats two goals consistent with the impression of data transformation as ‘denoising’: (1) preserving the highest-magnitude, most-evident relationships between nascent and mature RNA and (2) enhancing the less-prominent relationships. To that end, we quantify the effect of transformations on absolute and relative correlations between nascent and mature RNA. Although this moment-based measure is relatively simplistic, it is in line with the broad range of popular methods that directly manipulate correlation matrices to identify gene interactions and modules^{148,149,152}.

We find that nascent–mature correlations (Supplementary Fig. 16) are substantially distorted by transformations, with the sign of the correlation being frequently inverted by size normalization, PCA and UMAP. In addition, their magnitude relative to the mature–mature correlation distribution is generally not preserved, whether considering (Supplementary Fig. 17) or omitting (Supplementary Fig. 18) the signs.

Additionally, given Monod fits, we can directly compute biological correlations within a particular cell population. These correlations immediately follow from results in the Supplementary Methods (section S1.4). We observe that the correlations follow the expected trend: the ‘underlying’ correlations, before corruption by technical noise, are higher than observed correlations (Supplementary Fig. 19).

Dataset processing. Preprocessing: 10x PBMCs and Allen GABAergic neurons. We obtained raw 10x v2 and v3 FASTQ files from the 10x Genomics website for the 1k PBMC samples^{153,154}. For the GABAergic cells, we use the cell types (*Vip-*, *Sst-*, *Lamp5-* and *Pvalb-* expressing neurons) from the B08 sample generated by the Allen Institute for Brain Science^{56,136}, for which we previously performed the fits²⁴. We quantified the datasets using kallisto|bustools 0.26.0 with a prebuilt GRCh38 or mm10 genome from <https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest> (2020-A version), as described in ‘Overview of Monod inference’.

Filtering. For the PBMC datasets, we used the precomputed 10x Genomics *K*-means clustering results with *K* = 4, assigning cluster 1 to T cells, cluster 2 to monocytes, cluster 3 to B cells and cluster 4 to erythrocytes and low-quality cells, determining these assignments by manual curation. We also selected cells with more than 1,000 or 10,000 unique molecule identifiers in total (in the v2 and v3 data, respectively). We then selected for the same genes across both 10x v2 and v3 PBMC datasets with moderate to high expression using Monod, as described in ‘Parameter inference procedure’. From genes that passed these thresholds, we randomly selected 3,500 genes to fit.

Inference and analysis of biophysical parameters. For the PBMC datasets, based on previous technical sampling parameters fits described by Gorin and Pachter²⁴, we set $\{\log_{10}(C_N) = -6.105263157894737, \log_{10}(\lambda_M) = -1.125\}$ for inference with Monod. We fit the bursty model of transcription with (Poisson) length-biased sampling, as was done with the Allen GABAergic data, to each cell type. We then applied χ^2 testing and Hellinger distance criteria as in ‘Rejection statistics and parameter uncertainty’ to reject genes with poor fits.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets released by the Allen Institute for Brain Science were downloaded from http://data.nemoarchive.org/biccn/grant/u19_zeng/zeng/transcriptome/scell/10x_v3/mouse/raw/MOp/ and filtered according to the metadata annotations at http://data.nemoarchive.org/biccn/grant/u19_zeng/zeng/transcriptome/scell/10x_v3/mouse/processed/analysis/10x_cells_v3_AIBS/ refs. 56,136,155. The paired single-cell and single-nucleus mouse brain datasets as well as the human PBMC datasets were obtained from the 10x Genomics website. Control and IdU-perturbed mESC data⁴⁷ were obtained from GSE176044 at the Gene Expression Omnibus. The mouse germ cell dataset⁵⁷ was obtained from Gene Expression Omnibus repository GSE136220, the patient-derived PDAC tumor sample dataset⁶¹ was from GSE202051, and intestinal radiation therapy data⁶⁰ were from GSE165318. Single-nucleus mESC data¹⁰⁸ were obtained using Sequence Read Archive run accession number SRR18364193. Fluorescent intensity values for seqFISH+ experiments on mESCs¹⁰⁷ were obtained from

Zenodo (package 7693825)¹⁵⁶. The prebuilt GRCh38 or mm10 genome from <https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest> (2020-A version) was used for quantification of datasets. All loom files with nascent and mature count matrices for the datasets used in this study and all Monod fits have been deposited in Zenodo (package 15051840)^{157,158}.

Code availability

Notebooks that reproduce all the filtering, fitting and analysis procedures have been deposited also in Zenodo (package 15051840)¹⁵⁷ and are available at https://github.com/pachterlab/monod_examples/tree/main/manuscript_computation. This GitHub repository also contains a Google Colaboratory notebook that illustrates the Monod workflow with a small dataset. The Monod software used for all analysis is available as a pip installable package with an API available at <https://monod-examples.readthedocs.io>.

References

- Harris, C. R. et al. Array programming with NumPy. *Nature* **585**, 357–362 (2020).
- Virtanen, P. et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **17**, 261–272 (2020).
- Brodtkorb, P. A. & D’Errico, J. numdifftools <https://pypi.org/project/numdifftools/> (2021).
- Melsted, P. et al. Modular, efficient and constant-memory single-cell RNA-seq preprocessing. *Nat. Biotechnol.* **39**, 813–818 (2021).
- Gorin, F. & Pachter, L. Intrinsic and extrinsic noise are distinguishable in a synthesis – export – degradation model of mRNA production. Preprint at bioRxiv <https://doi.org/10.1101/2020.09.25.312868> (2020).
- Ham, L., Schnoerr, D., Brackston, R. D. & Stumpf, M. P. H. Exactly solvable models of stochastic gene expression. *J. Chem. Phys.* **152**, 144106 (2020).
- Ham, L., Jackson, M. & Stumpf, M. P. H. Pathway dynamics can delineate the sources of transcriptional noise in gene expression. *eLife* **10**, e69324 (2021).
- Jiang, Q. et al. Neural network aided approximation and parameter inference of non-Markovian models of gene expression. *Nat. Commun.* **12**, 2618 (2021).
- The MathWorks. MATLAB R2022a Symbolic Math Toolbox (MathWorks, 2022).
- The MathWorks. MATLAB R2022a (MathWorks, 2022).
- Tang, W., Jørgensen, A. C. S., Marguerat, S., Thomas, P. & Shahrezaei, V. Modelling capture efficiency of single cell RNA-sequencing data improves inference of transcriptome-wide burst kinetics. *Bioinformatics* **39**, btad395 (2023).
- Munsky, B., Li, G., Fox, Z. R., Shepherd, D. P. & Neuert, G. Distribution shapes govern the discovery of predictive models for gene regulation. *Proc. Natl Acad. Sci. USA* **115**, 7533–7538 (2018).
- Allen Institute for Brain Science. FASTQ Files for Allen v3 Mouse MOp Samples http://data.nemoarchive.org/biccn/grant/u19_zeng/zeng/transcriptome/scell/10x_v3/mouse/raw/MOp (2020).
- Melsted, P. áll., Ntranos, V. & Pachter, L. The barcode, UMI, set format and BUStools. *Bioinformatics* **35**, 4472–4473 (2019).
- Burnham, K. P. & Anderson, D. R. *Model Selection and Multimodel Inference: a Practical Information–Theoretic Approach* 2nd edn (Springer, 2002).
- Ly, A., Marsman, M., Verhagen, J., Grasman, R. P. P. & Wagenmakers, E.-J. A tutorial on Fisher information. *J. Math. Psychol.* **80**, 40–55 (2017).
- Lopez, R., Regier, J., Cole, M. B., Jordan, M. I. & Yosef, N. Deep generative modeling for single-cell transcriptomics. *Nat. Methods* **15**, 1053–1058 (2018).

141. Halpern, K. B. et al. Nuclear retention of mRNA in mammalian tissues. *Cell Rep.* **13**, 2653–2662 (2015).
142. McKellar, D. W. et al. Large-scale integration of single-cell transcriptomic data captures transitional progenitor states in mouse skeletal muscle regeneration. *Commun. Biol.* **4**, 1280 (2021).
143. Jew, B. et al. Accurate estimation of cell composition in bulk expression through robust integration of single-cell information. *Nat. Commun.* **11**, 1971 (2020).
144. Pedregosa, F. et al. Scikit-learn: machine learning in Python. *J. Mach. Learn. Res.* **12**, 2825–2830 (2011).
145. La Manno, G. et al. RNA velocity of single cells. *Nature* **560**, 494–498 (2018).
146. Montserrat Ayuso, T. & Esteve-Codina, A. Revealing the prevalence of suboptimal cells and organs in reference cell atlases: an imperative for enhanced quality control. *BMC Genomics* **25**, 1124 (2024).
147. Clarke, Z. A. & Bader, G. *MALAT1* expression indicates cell quality in single-cell RNA sequencing data. Preprint at *bioRxiv* <https://doi.org/10.1101/2024.07.14.603469> (2024).
148. Pratapa, A., Jaliha, A. P., Law, J. N., Bharadwaj, A. & Murali, T. M. Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nat. Methods* **17**, 147–154 (2020).
149. Kim, D. et al. Gene regulatory network reconstruction: harnessing the power of single-cell multi-omic data. *NPJ Syst. Biol. Appl.* **9**, 51 (2023).
150. Stumpf, M. P. H. Statistical and computational challenges for whole cell modelling. *Curr. Opin. Syst. Biol.* **26**, 58–63 (2021).
151. Saint-Antoine, M. M. & Singh, A. Network inference in systems biology: recent developments, challenges, and applications. *Curr. Opin. Biotechnol.* **63**, 89–98 (2020).
152. Stumpf, M. P. H. Inferring better gene regulation networks from single-cell data. *Curr. Opin. Syst. Biol.* **27**, 100342 (2021).
153. 10x Genomics. *1k PBMCs from a Healthy Donor (v2 Chemistry), Single Cell Gene Expression Dataset by Cell Ranger 3.0.0* <https://www.10xgenomics.com/datasets/1-k-pbm-cs-from-a-healthy-donor-v2-chemistry-3-standard-3-0-0> (2018).
154. 10x Genomics. *1k PBMCs from a Healthy Donor (v3 Chemistry), Single Cell Gene Expression Dataset by Cell Ranger 3.0.0* <https://www.10xgenomics.com/datasets/1-k-pbm-cs-from-a-healthy-donor-v3-chemistry-3-standard-3-0-0> (2018).
155. Allen Institute for Brain Science. *Analyses for Allen v3 Mouse MOF Samples* http://data.nemoarchive.org/biccn/grant/u19_zeng/zeng/transcriptome/scell/10x_v3/mouse/processed/analysis/10X_cells_v3_AIBS/ (2020).
156. Takei, Y., Yang, Y. & Cai, L. High-resolution spatial multi-omics datasets. *Zenodo* <https://doi.org/10.5281/zenodo.7693825> (2023).
157. Gorin, G., Pachter, L., Chari, T., Carilli, M. & Vastola, J. Monod supporting data. *Zenodo* <https://doi.org/10.5281/zenodo.15051840> (2025).
158. Jayakumar, K. & Mundassery, D. A. On Moran's bivariate gamma and bivariate negative binomial distribution. *Calcutta Stat. Assoc. Bull.* **59**, 15–28 (2007).

Acknowledgements

We thank S. Boeshaghi and M. Fang for useful discussions in the course of developing Monod. G.G. and L.P. were partially funded by NIH U19MH114830 and NIH 5UM1HG012077-02. M.C. is supported by the National Science Foundation Graduate Research Fellowship Program under grant no. 2139433. T.C. was supported by the Eric and Wendy Schmidt Center at the Broad Institute of MIT and Harvard. Part of this work was performed during G.G.'s Data Sciences Co-op with Celsius Therapeutics.

Author contributions

All authors contributed extensively to the work presented in this paper. The first draft of the paper was conceptualized and written by G.G. and L.P. 'Assumptions and approach of Monod' was predominantly conceptualized by G.G., J.J.V. and L.P. 'Monod generalizes differential expression analyses' was predominantly conceptualized and executed by G.G., T.C. and L.P. 'Monod identifies strategies of resistance and recovery' and 'Model selection and insight about gene regulation strategies' were predominantly conceptualized and executed by M.C. and L.P. 'Principled integration of multiple modalities' was predominantly conceptualized and executed by G.G., J.J.V., M.C. and L.P. 'Assessing loss of biological signal after preprocessing' was predominantly conceptualized and executed by G.G., J.J.V. and L.P.

Competing interests

G.G. is an employee of Fauna Bio. The remaining authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41592-025-02832-x>.

Correspondence and requests for materials should be addressed to Lior Pachter.

Peer review information *Nature Methods* thanks the anonymous reviewers for their contribution to the peer review of this work. Primary Handling Editor: Lin Tang, in collaboration with the *Nature Methods* team.

Reprints and permissions information is available at www.nature.com/reprints.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No software was used for data collection.
Data analysis	The open-source kallisto bustools software (v0.29.0 and v0.26.0) was used to process the collected data. The Monod 0.2.6.0 package was used to perform inference and analysis of this processed data (https://monod-examples.readthedocs.io/en/latest/). Both softwares are available as pip installable packages. For analysis of Monod results SciPy v1.13.0 , scikit-learn v1.0.1, NumPy v1.26.4, and numdifftools v0.9.41 were used.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The datasets released by the Allen Institute for Brain Science were downloaded from <http://data>.

nemoarchive.org/biccn/grant/u19_zeng/zeng/transcriptome/scell/10x_v3/mouse/raw/MOp/, and filtered according to the metadata annotations at http://data.nemoarchive.org/biccn/grant/u19_zeng/zeng/transcriptome/scell/10x_v3/mouse/processed/analysis/10X_cells_v3_AIBS/ [57, 128, 129]. The paired single-cell and single-nucleus mouse brain datasets, as well as the human PBMC datasets, were obtained from the 10x Genomics website. The control and IdU-perturbed mouse embryonic stem cell (mESC) data from the study by [49] was obtained from GSE176044 at the Gene Expression Omnibus (GEO). The mouse germ cell dataset from [32] was obtained from GEO repository GSE136220, the patient-derived PDAC tumor samples [59] from GSE202051, and the intestinal radiation therapy data [61] from GSE165318. The single-nucleus mESC data [112] was obtained using the Sequence Read Archive run accession number SRR18364193. Fluorescent intensity values for seqFISH+ experiments on mESCs [111] were obtained from Zenodo package 7693825. The pre-built GRCh38 or mm10 genome from <https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest> (2020-A version) was used for quantification of datasets. All loom files with nascent and mature count matrices for the datasets used in this study and all Monod fits have been deposited as the Zenodo package 15051840.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

n/a

Reporting on race, ethnicity, or other socially relevant groupings

n/a

Population characteristics

n/a

Recruitment

n/a

Ethics oversight

n/a

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences

☐ Behavioural & social sciences

☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

No sample size calculation was performed. We selected 9 previously published datasets which covered a range of sizes (cell numbers), experimental settings, cell types, and biological models for analysis. Thus we captured datasets with both biological and technical range to demonstrate the use of the proposed method, Monod, across data scales (from hundreds to tens of thousands of cells) and for use in biological hypothesis development in different biological systems (in vitro models and more complex in vivo systems in multiple species).

Data exclusions

Only data from cells previously determined as cells from the original studies were used in analysis, to avoid use of noise/data artifacts and to use previously obtained labels or annotations of the cells.

Replication

All experimental findings used were previously published sources and thus relied on the availability of replicates in these datasets. Biological replicates were used for analysis when available to validate the parameters inferred by our analysis, demonstrating consistent parameter inference across replicates. For datasets where biological replicates were not available, as is commonly the case in single-cell genomics, uncertainty analysis of the inferences from these single samples was performed to demonstrate the capacity of Monod to still propose novel biological hypotheses while providing a measure of confidence in these outcomes.

Randomization

Random sample allocation was not necessary in this study. All inferences made in the analysis are performed independently of each other, and have no effect on the outcome of other analyses.

Blinding

Blinding was not necessary in this study as the goal of the study was to demonstrate the use of Monod in revealing new biological hypotheses within previously categorized cells/groups. Thus the relevant labels or partitions of the data were used as a part of the analysis, to investigate the parameters inferred within these labeled groups. The resulting inferences are then compared to previous statements and analyses made within these groups in the prior studies.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

n/a

Novel plant genotypes

n/a

Authentication

n/a